

Equal Before the Measure

Shared Measurement and Perceived Fairness in an Unequal Order

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Abstract: Shared measurement can make an unequal opportunity order appear fairer even when unequal access remains explicit. I distinguish the scope of allocation, which determines who competes with whom, from the scope of measurement, which determines who faces a common evaluative instrument. Candidates in separate allocation pools can thus share a limited equality before the measure without competing directly. A quasi-experimental analysis of China's mid-2000s college entrance examination reform links the transition from shared to province-set papers with lower perceived social fairness years later, while admissions remained province-based. A preregistered survey experiment on a hypothetical national talent-selection program likewise finds lower perceived social fairness under province-specific papers, holding allocation, within-province selection rules, and paper-setting authority constant. The findings identify measurement scope as a distinct dimension of institutional design and shared measurement as a potential basis for legitimating unequal opportunity orders without revising their distributive bargains.

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Introduction

In a typical year, at 9:00 a.m. Beijing time on June 7, more than ten million candidates across China begin the Chinese-language portion of the *gaokao*, the National College Entrance Examination. The examination follows a single national timetable, and across much of the country candidates take one of a small number of centrally written papers, each shared across several provinces.¹ Local authorities adhere to that timetable despite conditions ranging from extreme heat to monsoon rains and flooding. Yet this nationally coordinated evaluative moment sits atop a territorially fragmented system of distribution. University admission quotas are assigned province by province, cutoff scores vary across provinces, and both are publicly announced. Eligibility for a provincial pool is ordinarily tied to household registration and schooling history, and cross-provincial switching is tightly regulated.² Candidates thus face common evaluative tasks while competing primarily within separate provincial pools.

The scope of shared measurement, however, has not remained fixed. Beginning in 2004, the use of province-set *gaokao* papers expanded sharply, reaching sixteen provincial-level jurisdictions by 2006, while province-based admissions and the national examination timetable remained in place. In 2014, the central government reversed direction. By 2016, twenty-six jurisdictions used nationally written papers for all or part of the examination, although admissions remained province-based. Officials cited uneven item quality and the costs of producing numerous paper variants as

1. “National papers” do not refer to a single paper used throughout China. The national examination authority prepares several centrally written variants, each used by multiple provinces ([General Office of the Ministry of Education of the People’s Republic of China 2018](#); [Ministry of Education of the People’s Republic of China 2016](#); Zhang and Qiu 2016).

2. The Ministry of Education’s “Sunshine Project” requires disclosure of admission policies and plans, provincial control scores, admission procedures, and results ([Ministry of Education of the People’s Republic of China 2005](#)). Eligibility is not determined exclusively by household registration. Some children of migrant families may take the examination in their destination province. Provincial authorities nevertheless regulate access using combinations of household registration, student registration, and actual schooling history ([Ministry of Education of the People’s Republic of China 2018](#); [Ministry of Education of the People’s Republic of China and Ministry of Public Security of the People’s Republic of China 2016](#)).

reasons for the later reversal.³ These considerations help explain the administrative stakes, but the changes also raise a question about public evaluation. What happens to perceived fairness when candidates lose a common measure while remaining in the same unequal allocation pools?

The persistence of unequal allocations does not make the answer obvious. Research on distributive attitudes in China shows that substantial inequality need not translate into correspondingly strong perceptions of injustice (Whyte 2010). If fairness judgments depended on allocation alone, changing the paper arrangement would make no difference. A shared measure, however, also makes unequal returns to similar performance easier to see. If relief from such comparisons dominated, withdrawing the common paper would make the unequal arrangement appear fairer. Yet people may also value consistent evaluative treatment even when allocation remains unequal, as procedural justice research suggests (Gilliland 1993; Leventhal 1980).

I argue that shared measurement can make an unequal opportunity order appear fairer without changing how opportunities are allocated. I distinguish the *scope of allocation*, which determines who competes with whom and how opportunities are divided among pools, from the *scope of measurement*, which determines who faces a common evaluative instrument. Candidates competing separately for the same class of opportunity can thus share a limited *equality before the measure* despite unequal preparation, selection chances, and returns. When the common task is regarded as credible and appropriate, candidates and observers may value this shared evaluative standing, sustaining an *evaluative fairness premium*. Where unequal allocations are already publicly known, withdrawing the common instrument may leave awareness of disadvantage intact while removing that standing and weakening the public basis for contesting unequal returns. I expect this loss to be especially consequential for disadvantaged candidates when a common basis for seeking recognition of comparable performance is itself valued as part of fair treatment. Because evaluative treatment in high-stakes selection can inform judgments of how society recognizes performance and distributes opportunity, these losses may also make the wider social order appear less fair.

3. At a 2016 press conference, Minister of Education Yuan Guiren described uneven quality across the seventeen paper-setting units as the largest problem and also pointed to the higher cost of province-level paper-setting (Zhang and Qiu 2016).

I test this argument with two complementary studies. Study 1 examines the staggered 2004–2006 *gaokao* reform, when fourteen provincial-level jurisdictions adopted province-set papers while comparison jurisdictions retained national papers. I use eight waves of the Chinese General Social Survey (CGSS), fielded between 2010 and 2021, to examine general assessments of social fairness recorded years after the reform. Potential exposure is assigned using respondents’ expected *gaokao* years and their inferred hukou provinces near examination age, drawing on birth, household-registration, and migration information. The preferred sample comprises 3,387 respondents in the 2001–2007 expected-examination cohorts whose upper-secondary education or higher makes engagement with *gaokao* preparation or participation plausible.

Cohort-based event-study estimates link the reform to lower perceived social fairness measured years later. The adoption-boundary estimate is $-.21$ points on the five-point scale (approximately $-.21$ SD), averaged across the three cohorts nearest the institutional transition—the final national-paper cohort and the first two province-set-paper cohorts. An exploratory comparison among jurisdictions adopting in 2004 finds a sharper decline where undergraduate institutions were scarcer relative to the number of *gaokao* candidates before the reform. With relative institutional supply serving as a proxy for provincial opportunity conditions, this pattern is consistent with greater fairness costs from withdrawing common evaluative terms where opportunities for higher education were more constrained.

The historical reform simultaneously narrowed shared measurement and transferred paper-setting authority from the center to the provinces. Study 2 separates these institutional changes through a preregistered survey experiment among adults in mainland China ($N = 2,413$). Respondents evaluate a hypothetical national talent-selection program in which opportunities are allocated unequally across provinces and applicants compete within provincial pools. The vignettes vary whether applicants nationwide take a common paper or province-specific papers, while the same administrative body prepares all papers in both conditions. Provincial allocation and within-province selection rules are also held constant in the written vignettes. Fairness ratings of a society that uses the system for talent selection are lower by $.15$ points on the five-point scale (Glass’s $\delta = -.14$) in

the province-specific-paper condition, and the selection system itself is also judged less fair. The experiment extends the institutional comparison beyond university admission and shows that the fairness response occurs even when unequal opportunities remain explicit and the stated allocation rules are unchanged.

These findings identify measurement scope as a distinct dimension of meritocratic institutional design. Existing research examines how institutions define merit and justify social advantage (Acominotti and Sauder 2026; J. J. B. Mijs 2016), how access to competition shapes perceived mobility and redistributive preferences (Liu 2024), and how examination outcomes affect evaluations of the selection process and national identification (Kuipers 2023). Research on quantification shows that common metrics construct relations of comparability and can become resources for both governance and contestation (Espeland and Stevens 1998; Mennicken and Espeland 2019). This paper shows that the fairness implications of meritocratic selection extend beyond the treatment of candidates who compete directly with one another. Shared measurement across separate allocation pools can make the wider opportunity order appear fairer even when unequal access and returns remain explicit.

This paper also contributes to research on institutional design under authoritarian rule. Institutions facilitate co-optation and policy compromise (Gandhi and Przeworski 2007) and can sustain power-sharing by addressing monitoring and commitment problems (Boix and Svulik 2013; Meng, Paine, and Powell 2023). Consultation and legal aid also influence how citizens evaluate public authorities (Malesky and Taussig 2019; Truex 2017; Whiting 2017). I extend this state–society perspective by showing that measurement scope can shape the perceived fairness of an allocation order without changing its distributive bargain. Because perceived fairness can provide a basis for institutional legitimacy (Levi, Sacks, and Tyler 2009), shared measurement may offer a legitimating resource where group allocations are politically difficult to revise.

Shared Measurement and Perceived Fairness in an Unequal Order

Two Boundaries of Meritocratic Selection

High-stakes selection often allocates opportunities across group-specific pools and then ranks candidates within each pool. University places may be apportioned across regions, public positions across jurisdictions, and benefits across status categories. These arrangements can reflect historical settlements or political bargains that become costly to revise once institutionalized (Pierson 2000; Tilly 1998). Meritocratic competition therefore often operates within an existing group-based allocation of opportunities.

I distinguish the *scope of allocation* from the *scope of shared measurement*. Allocation scope determines who competes with whom and the share of opportunities assigned to each pool; measurement scope determines who faces the same concrete evaluative instrument. These boundaries need not coincide. Candidates may face a common task while remaining in separate pools with different quotas, cutoffs, and within-pool rankings. Conversely, an institution may introduce pool-specific instruments without changing who competes with whom or how opportunities are formally allocated.

I call the use of a common instrument across separate allocation pools *evaluative integration without allocative integration*. Equality in one dimension need not imply equality in others (Sen 1979). Common evaluative terms can coexist with inequalities in preparation, access, selection chances, and outcomes. Measurement scope is also distinct from administrative centralization. A single authority may produce one instrument or several, while separate authorities may coordinate around a common one.

Cross-pool evaluation is especially consequential when candidates belong to a common opportunity order. The argument applies most clearly when pools form part of the same institutional system, distribute the same class of opportunity, and cannot be readily chosen or switched by candidates. Within such an order, candidates remain claimants to the same class of opportunity even

when they do not compete directly. Replacing a shared instrument with pool-specific ones extends an existing group division into measurement. Group membership then determines both the opportunities for which candidates compete and the instrument through which their performance is assessed.

Shared Measurement as Publicly Legible Evaluative Parity

A shared evaluative instrument provides a publicly legible basis for comparing performance across separate allocation pools. Commensuration does not merely record preexisting differences; it creates relations in which similarity and relative performance can be expressed on a common metric (Espeland and Stevens 1998). Quantification can lend public authority to these judgments by presenting them as governed by impersonal rules rather than personal discretion (Porter 1996). Measurement scope, in turn, shapes how far this shared basis for comparison and judgment extends across group boundaries (Buitron, Mühlfried, and Steinmüller 2026).

I call this limited parity *equality before the measure*. It exists when candidates in a common opportunity order face common evaluative terms despite competing in separate allocation pools. Consistent with a relational understanding of equality (Anderson 1999), the concept concerns candidates' standing in relation to one another under evaluation rather than the distribution of opportunities.

This parity does not imply that a common instrument is more valid or less biased. Distinct instruments can be psychometrically equated, while a single instrument may function differently across populations (Kolen and Brennan 2014). The relevant distinction is between directly visible commonality and institutionally certified equivalence. A shared instrument makes the common task apparent, whereas observers must rely on specialized procedures or institutional assurances to recognize equivalence across separate instruments. Public legibility concerns this basis for comparison, not disclosure of examination content. Shared measurement thus lowers the evidentiary burden of cross-pool comparison without establishing the fairness of the instrument or its associated returns.

The argument about common evaluative standing builds on procedural justice research. Con-

sistency in evaluative treatment is a central fairness criterion (Gilliland 1993; Leventhal 1980), while the relational model emphasizes what treatment conveys about individuals' standing within a group beyond the outcomes they receive (Tyler and Lind 1992). Both shared and pool-specific instruments can preserve consistent treatment within pools. A common instrument additionally places candidates across pools on visibly common evaluative terms, even when they do not compete directly. This common evaluative standing may therefore be valued by candidates across pool positions and by observers without a direct stake in selection.

For candidates in disadvantaged pools, where comparable measured performance receives less favorable returns, a shared task also supplies a common language for contesting unequal returns (Boltanski and Thévenot 2006). Candidates can invoke equal or stronger performance under the institution's own standard to argue that their poorer opportunities reflect pool membership rather than weaker performance. Research on rightful resistance similarly shows how officially endorsed principles and commitments can support challenges to authorities' practices (O'Brien and Li 2006). A common measure makes such claims intelligible beyond the pool even when they produce no change in allocation.

With formal allocation unchanged, withdrawing the shared measure therefore entails a relational loss. Disadvantaged candidates retain their allocation disadvantage while losing the visibly common evaluative terms that supported their claims. For candidates in advantaged pools, where comparable performance receives more favorable returns, the same change may furnish an *incomparability alibi*. Differences between instruments provide institutional grounds for discounting cross-pool comparisons, making favorable returns less readily contestable.

From Evaluative Parity to Perceived Social Fairness

Shared measurement can therefore generate an *evaluative fairness premium*, whereby an otherwise unequal selection arrangement appears fairer under a common measure than under pool-specific instruments. Candidates and observers may value common evaluative standing while regarding the allocation of opportunities as unfair. This premium requires neither overlooking unequal opportunities nor treating common evaluation as compensation for them.

Comparability can also work in the opposite direction by exposing unequal returns to similar measured performance. Sharper comparison can increase dissatisfaction, and information about inequality can weaken meritocratic accounts of success (Card et al. 2012; Mijs and Hoy 2022). If relief from unfavorable comparisons dominated, withdrawing the common measure would improve rather than worsen perceived fairness.

The balance depends partly on what comparison newly reveals. Where unequal allocations are already publicly known, withdrawing a common measure may leave awareness of disadvantage intact while removing shared evaluative standing. When the common task is also regarded as credible, relevant, and appropriate, I expect this loss to outweigh, on average, both relief from unfavorable comparison and the benefits of shielding favorable returns. The premium may weaken or reverse when differentiated instruments are understood as necessary accommodations, the common measure is regarded as biased, or local autonomy is valued over common evaluative treatment (Buitron, Mühlfried, and Steinmüller 2026).

Pool position also shapes this balance. For disadvantaged candidates, withdrawal weakens the shared public basis for seeking recognition of comparable performance and contesting unequal returns. When the ability to substantiate such claims is regarded as part of fair treatment, this loss can lower perceived fairness while leaving their allocation disadvantage unchanged. Where that disadvantage is already apparent, I expect this additional fairness cost to outweigh any greater relief from comparison that disadvantaged candidates experience relative to advantaged ones. For advantaged candidates, the *incomparability alibi* may partly offset the loss of common evaluative standing. Withdrawal should therefore impose a greater fairness cost on disadvantaged candidates, without necessarily improving fairness judgments among advantaged candidates.

Fairness judgments about selection may also extend to the wider social order. High-stakes selection institutions organize access to consequential opportunities and provide public standards for assessing performance. Candidates and observers may therefore interpret evaluative treatment within these institutions as indicative of how society recognizes performance and distributes opportunity. Experiences with selective admission can revise beliefs about whether success reflects

effort, luck, or structural advantage, although changes in general beliefs about societal inequality may be smaller (Wetter and Finger 2023). Beliefs about school meritocracy can likewise inform judgments of social and income inequality (Batruch et al. 2023).

Such judgments may remain relevant after selection. Beliefs about meritocracy and structural inequality continue to evolve during college and as students develop understandings of future career opportunities (J. J. Mijs 2023; Tang, Li, and Wu 2026). An earlier evaluative arrangement may remain one input into later assessments of social fairness, even as subsequent experiences reinforce, attenuate, or revise its meaning.

For high-stakes systems in which unequal group allocations are publicly known and the common task is regarded as credible and appropriate, I expect the following.

Hypothesis 1. Within a common opportunity order, perceived social fairness will be lower on average under pool-specific measurement than under shared measurement, holding formal allocation and within-pool selection rules constant.

The Political Stakes of Measurement Scope under Authoritarian Rule

The evaluative fairness premium has political significance where group-based allocations are difficult to revise. Reallocation can impose identifiable losses and unsettle established political bargains. Research on redistribution highlights the importance of elite alignments and the potential tradeoffs between short-term political support and longer-term authoritarian survival (Albertus 2015; Wallace 2013). Measurement scope offers a distinct institutional margin on which evaluative terms can change without altering formal group shares, redrawing allocation pools, or changing within-pool selection rules.

Such institutional choices matter because procedural justice can contribute to citizens' willingness to defer to governmental authorities (Levi, Sacks, and Tyler 2009). Under authoritarian rule, legitimation constitutes a pillar of stability alongside repression and co-optation (Gerschewski 2013). In single-party Vietnam, offering firms the opportunity to comment on draft regulations improved their evaluations of regulatory authority and increased compliance, even when they sub-

mitted no substantive comments (Malesky and Taussig 2019). The perceived integrity of selection can also contribute to acceptance. Perry (2015) links the *gaokao*'s legitimating appeal to blind grading and its reputation for incorruptibility despite highly selective admission. Access to competition can likewise shape political attitudes. Eligibility for China's standardized civil-service examination increased perceived upward mobility and weakened demand for redistribution among college-educated citizens (Liu 2024).

Measurement scope adds to these accounts by directing attention to common evaluative treatment across allocation boundaries. Because perceptions of fair treatment can support institutional legitimacy (Levi, Sacks, and Tyler 2009; Mazepus and van Leeuwen 2020; Walters and Bolger 2019), shared measurement may furnish authorities with a resource for justifying an unequal opportunity order even when its inequalities remain publicly visible. Narrowing measurement scope may weaken that resource, regardless of whether legitimation motivated the institutional change.

The empirical analysis pairs real-world institutional variation with an experimental comparison. Study 1 estimates the reduced-form effect of China's mid-2000s transition to province-set *gaokao* papers on social fairness judgments recorded years later. Study 2 examines the shared-measurement proposition through a preregistered survey experiment involving a hypothetical national talent-selection program, holding allocation, within-province selection rules, and paper-setting authority constant in the vignettes.

Study 1: The Mid-2000s Expansion of Province-Set Gaokao Papers

Institutional Context, Reform, and Cohort Exposure

Restored in 1977, the *gaokao* became a mass institution as higher education expanded after 1999. It serves as China's principal route into higher education and, through it, to upward mobility, elite careers, and social recognition (Howlett 2022b; Jia, Li, and Cousineau 2025). Beyond university selection, the examination presents advancement as the outcome of an impersonal, rule-bound contest rather than inherited privilege or overt political favoritism (Howlett 2022b; Jiang 2024).

Candidates from less advantaged backgrounds often regard it as one of the few legitimate routes for “changing fate” (Howlett 2022b). Its significance as a rite of passage can outlast the academic content it tests (Howlett 2022a), making changes in its organization potentially relevant to later judgments of social fairness.

Before the broad expansion of province-set papers, most provinces used centrally written papers shared across provincial boundaries, while admissions remained organized around provincial quotas and cutoff scores. This tension became especially visible in 2001, when three recent high-school graduates from Qingdao traveled to Beijing seeking to file administrative suits against the Ministry of Education. They challenged substantially lower admission thresholds for Beijing candidates despite participation in the same national examination.⁴ The episode does not establish the origins of the subsequent reform. It nevertheless illustrates how shared measurement made unequal returns to examination performance publicly visible and contestable.

Beijing adopted province-set papers in 2002, following Shanghai’s earlier adoption in 1985. In 2003, the theft of national *gaokao* papers from the education bureau of Nanbu County exposed a security vulnerability of shared papers: a local breach could jeopardize examinations elsewhere.⁵ Between 2004 and 2006, fourteen additional provincial-level jurisdictions adopted province-set papers, bringing the total to sixteen. The reform replaced nationally shared papers in some or all examination subjects, narrowing shared measurement while decentralizing paper-setting authority. In announcing the 2004 expansion, officials stated that admission policies, enrollment-planning arrangements, and the national examination timetable would remain unchanged.⁶ Figure 1 sum-

4. Li Yan, “三考生今向高法递诉状，能否被受理尚难确定” [“Three Candidates to Submit Administrative Complaints to the Supreme People’s Court; Whether the Cases Will Be Accepted Remains Uncertain”], *Qilu Wanbao* [*Qilu Evening News*], August 23, 2001, republished by Sina News, <https://news.sina.com.cn/c/2001-08-23/337887.html> (accessed August 30, 2026). The report states that the proposed litigation followed the publication of provincial admission cutoffs. For liberal-arts candidates, Shandong’s first-tier cutoff was 575, compared with 456 in Beijing.

5. China News Service, “四川高考试卷被盗案：窃贼因窃国家秘密罪判 7 年” [“Sichuan Gaokao Paper Theft Case: Thief Sentenced to Seven Years for Illegally Obtaining State Secrets”], August 26, 2003, <https://www.chinanews.com.cn/n/2003-08-26/26/339446.html> (accessed August 30, 2026).

6. Liu Haifeng, “高考命题从分到统的历史逻辑” [“The Historical Logic of *Gaokao* Paper-

marizes adoption timing, with province-level details reported in Table A.1 of the Supplementary Materials.

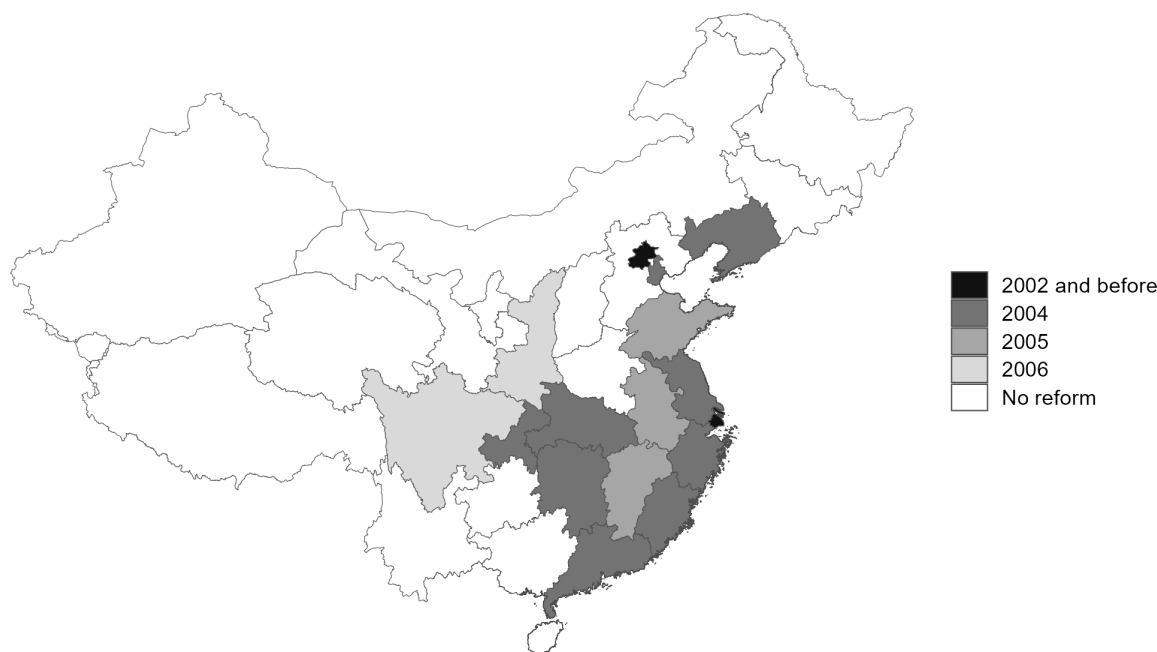


Figure 1: Staggered Provincial Adoption of Province-Set Gaokao Papers

Potential exposure to the reform could arise through its public announcement as well as through preparation for and participation in a province-set examination. In most adopting provinces, the final national-paper cohort had completed the examination by the time formal authorization was announced in the implementation year; where local notice came earlier, some were still preparing for it.⁷ The announcement did not change their examination paper, but this cohort occupied a

Setting from Decentralization to Unification”], *Zhongguo Jiaoyu Bao* [China Education Daily], March 12, 2015, hosted by the Ministry of Education, https://www.moe.gov.cn/jyb_xwfb/xw_zt/moe_357/jyzt_2015nztzl/lianghui/pinglun/202103/t20210329_523321.html (accessed August 30, 2026). See also Liu Wanyong, “2004 年高考新增九省市单独命题” [“Nine Additional Provinces and Municipalities to Set Their Own *Gaokao* Papers in 2004”], *Zhongguo Qingnian Bao* [China Youth Daily], February 20, 2004, https://zqb.cyol.com/content/2004-02/20/content_822634.htm (accessed August 30, 2026).

7. The 2004 expansion was announced in February, several months before that year’s examination. The subsequent waves were likewise announced before implementation. “山东安徽江西三省今年高考获准自主命题” [“Shandong, Anhui, and Jiangxi Authorized to Set Their Own *Gaokao* Papers in 2005”], *Jinghua Shibao* [Beijing Times], March 11, 2005, <https://gaokao.chsi.com.cn/gkx/x/sd/200503/20050311/4944.html> (accessed August 30, 2026); CCTV International, “2006 年高考

distinctive position: its immediate successor faced a different paper-setting arrangement, whereas earlier national-paper cohorts were followed by another cohort under the national-paper arrangement. This juxtaposition could make the reform especially salient in relation to their own recent examination experience. Shared measurement could thereby appear as a contingent administrative arrangement rather than an intrinsic feature of a “national” examination (Timmermans and Epstein 2010). The first province-set-paper cohort directly encountered the transition in preparation and examination, while the following cohort was already in upper-secondary school and approaching the examination under the new arrangement. These three cohorts provide a focused window on a transition whose significance could vary with its timing in candidates’ life courses (Elder 1998).

These experiences could remain relevant to later attitudes without requiring a continuously held fairness judgment. Experiences and public events from adolescence and early adulthood are disproportionately represented in later recall (Rubin, Rahhal, and Poon 1998; Schuman and Corning 2012). Research on mobilization and repression in China also documents attitudinal consequences decades after exposure (Desposato, Wang, and Wu 2021). Because broad evaluative judgments draw on considerations accessible at the moment of response (Tourangeau, Rips, and Rasinski 2000; Zaller and Feldman 1992), the contrast between shared and province-set papers could remain a reference point for later assessments of social fairness. Its meaning may evolve with subsequent experiences while continuing to inform judgments about whether advancement rests on common evaluative terms across provinces.

Data and Research Design

Study 1 uses the spatial-cohort variation generated by staggered adoption in 2004, 2005, and 2006 within a generalized difference-in-differences framework. Related research uses similar variation

新增四川陕西两个自主命题省份” [“Sichuan and Shaanxi Added as Province-Set-Paper Jurisdictions in 2006”], February 10, 2006, <https://finance.cctv.com/news/science/20060210/101068.shtml> (accessed August 30, 2026). Local notice could appear still earlier. Yin Xiaohua, “四川省明年高考自主命题, 今年仍使用全国试卷” [“Sichuan to Set Its Own *Gaokao* Papers in 2006; National Papers Retained in 2005”], *Sichuan Ribao* [*Sichuan Daily*], January 16, 2005, republished by Sina News, <https://news.sina.com.cn/e/2005-01-16/08024846350s.shtml> (accessed August 30, 2026).

to examine the later attitudinal consequences of educational and selection institutions in China (Cantoni et al. 2017; Liu 2024). For each adoption wave, I compare differences in later judgments of social fairness across expected *gaokao* cohorts in adopting jurisdictions with the corresponding cohort differences in jurisdictions that retained centrally prepared papers throughout the focal window. The primary estimand summarizes the average change in the adopter–comparison gap across the final national-paper cohort and the first two province-set-paper cohorts, relative to an earlier national-paper cohort. Because adoption was not random, identification requires that, absent the reform transition and conditional on the included fixed effects, perceived social fairness would have followed parallel local cohort trends in adopting and comparison jurisdictions.

The individual-level data come from eight waves of the Chinese General Social Survey (CGSS), fielded in 2010, 2011, 2012, 2013, 2015, 2017, 2018, and 2021. This national repeated cross-sectional survey records demographic characteristics, educational attainment, hukou and migration information, and social attitudes (Bian and Li 2012). Perceived social fairness is measured by the question, “Overall, do you think society today is fair or unfair?” Responses range from 1, “completely unfair,” to 5, “completely fair,” and I retain the original scale. All responses were collected after the reform. Event time therefore indexes expected examination cohorts, rather than repeated pre- and post-reform measurements of attitudes.

I assign expected *gaokao* years using birth year and, where available, birth month. Expected examination year equals birth year plus eighteen for respondents born from January through August and birth year plus nineteen for those born from September through December, reflecting the school-entry cutoff. When birth month is unavailable, I use birth year plus eighteen. This fallback applies throughout the 2017 and 2018 waves, whose public-use files do not report birth month. These assignments approximate ordinary school progression and cannot capture early or delayed entry, grade repetition, or examination retaking. Appendix E in the Supplementary Materials examines sensitivity to alternative expected-year assignments.

Separately, I infer the province-level hukou jurisdiction most likely to have governed each respondent near examination age. A fixed hierarchy prioritizes hukou evidence over residence-based

proxies, using the information available in each CGSS wave. Current hukou province receives priority when registration in the current locality dates to birth or a transfer before age sixteen; historical hukou reports provide the next sources. Residence-based proxies are used only when no usable hukou-based source is available. When current hukou province is not directly reported, survey province serves as its proxy only for respondents reporting locally registered hukou. Conflicting sources are resolved by the same priority rule. Appendix B in the Supplementary Materials documents the complete hierarchy and sample construction; Appendix E examines robustness to retaining only high-confidence assignments and excluding respondents whose usable sources identify different provinces.

I exclude Beijing and Shanghai, which adopted province-set papers before the reform waves examined here. This leaves fourteen adopting and fifteen eligible comparison jurisdictions before respondent-level restrictions (Table A.1 of the Supplementary Materials). Among respondents with valid expected-year, hukou-province, and survey-province assignments, I retain those in eligible jurisdictions whose expected examination years fall between 2001 and 2007, yielding 6,207 respondents. This is the narrowest fixed cohort window that includes, for all three adoption waves, the three cohorts nearest the transition and two earlier national-paper cohorts used for normalization and temporal localization.

The preferred sample includes respondents whose highest educational level reported at the time of the survey was secondary specialized school, vocational high school, technical school, general senior high school, or higher. These educational trajectories make engagement with *gaokao* preparation or participation more plausible than among respondents who did not proceed beyond junior high school. Of the resulting 3,393 respondents, six lack the perceived social fairness outcome, leaving 3,387 across 28 inferred *gaokao*-hukou provinces. These comprise all fourteen adopting jurisdictions and fourteen comparison jurisdictions, with Tibet absent from the preferred sample.

A mutually exclusive low-education placebo sample contains 2,801 respondents with junior high school education or less and nonmissing responses to the perceived social fairness item. Assigned to the same expected-examination cohorts, these respondents were of comparable age when

the reform occurred but were substantially less likely to have prepared for or taken the *gaokao*. They could nevertheless have encountered the reform through personal contacts or public discussion. The placebo analysis asks whether a similar event-time pattern appears among this less directly involved group.

I estimate adoption-wave-by-event-time effects jointly using the cohort-interacted event-study specification of Sun and Abraham (2021). Let $G = \{2004, 2005, 2006\}$ denote the adoption waves. The specification is

$$Y_{ipcrs} = \sum_{g \in G} \sum_{\substack{k \in \mathcal{K}_g \\ k \neq -2}} \gamma_{gk} A_{pg} \mathbf{1}\{c - g = k\} + \alpha_p + \delta_c + \lambda_{rs} + \varepsilon_{ipcrs}. \quad (1)$$

Here, Y_{ipcrs} denotes perceived social fairness for respondent i , with inferred *gaokao*-hukou province p , expected examination year c , survey province r , and CGSS wave s . The indicator A_{pg} equals one if province p first adopted province-set papers in year g , and zero otherwise. Eligible nonadopting jurisdictions have $A_{pg} = 0$ for every $g \in G$. The set $\mathcal{K}_g$ contains the event times supported for adoption wave g , and $\mathbf{1}\{c - g = k\}$ indicates that expected examination year c falls at event time k relative to that wave. The terms α_p , δ_c , and λ_{rs} denote fixed effects for inferred *gaokao*-hukou province, expected examination year, and survey province by CGSS wave, respectively; ε_{ipcrs} is the error term. Potential reform exposure is assigned using inferred hukou province and expected examination year, while λ_{rs} accounts for the survey-province-by-CGSS-wave context in which perceived social fairness was subsequently measured.⁸ The baseline specification includes no individual-level covariates and does not use CGSS sampling weights.

Event time is $k = c - g$. Among adopters, $k = -1$ denotes the final national-paper cohort, $k = 0$ the first province-set-paper cohort, and $k = +1$ the immediately following cohort. The institutional

8. The CGSS item asks about social fairness without directly eliciting support for political authorities, distinguishing it from the regime-assessment questions central to research on self-censorship (Carter, Carter, and Schick 2024; Shen and Truex 2021). The fixed effects absorb additive reporting differences that are constant within inferred *gaokao*-hukou provinces, common across jurisdictions within expected-examination cohorts, or shared within survey-province-by-wave contexts. A reporting-bias explanation of the estimated effect would therefore require residual adopter-comparison differences in reporting that vary across cohorts around the reform boundary.

boundary therefore lies between $k = -1$ and $k = 0$. To allow for the final national-paper cohort's potential retrospective response to the transition, I use $k = -2$, the nearest earlier national-paper cohort outside this three-cohort window, as the reference for each adoption wave. The corresponding interaction is omitted, although observations from that cohort remain in the estimation sample. This choice provides a local benchmark but does not establish that earlier cohorts were unaffected by the reform. Conditional on the fixed effects, γ_{gk} measures the change from $k = -2$ to k in the gap in perceived social fairness between jurisdictions in adoption wave g and the nonadopting jurisdictions. The nonadopting jurisdictions provide the comparison trajectory, while the omitted cohort provides the within-wave reference. Because perceived social fairness is observed only in later CGSS waves, the coefficient at $k = -3$ serves as an earlier-cohort localization diagnostic. I also report the estimate at $k = +2$, which lies outside the focal estimand and is supported only by the 2004 and 2005 adoption waves.

After estimating the wave-specific coefficients $\hat{\gamma}_{gk}$, I aggregate them using fixed reform-wave weights. In the preferred sample's omitted $k = -2$ cells, the numbers of respondents in adopting jurisdictions are 176, 67, and 51 for the 2004, 2005, and 2006 waves, respectively. Their shares define the base weights:

$$(\omega_{2004}, \omega_{2005}, \omega_{2006}) = \frac{1}{294}(176, 67, 51) \approx (.599, .228, .173). \quad (2)$$

Let $G_k = \{g \in G : k \in \mathcal{K}_g\}$ denote the adoption waves with support at event time k . For $k \neq -2$, the reported event-time estimate is

$$\hat{\beta}_k = \sum_{g \in G_k} \frac{\omega_g}{\sum_{h \in G_k} \omega_h} \hat{\gamma}_{gk}. \quad (3)$$

The denominator renormalizes the base weights over the waves in G_k . All three adoption waves support every event time from $k = -3$ through $k = +1$, so the same weights apply throughout this common-support interval. At $k = +2$, only the 2004 and 2005 waves have support, and their weights are renormalized to approximately .724 and .276. The low-education placebo analysis uses the same base weights to hold the reform-wave composition of the aggregated estimates constant across the

two education groups. Within the common-support interval, fixing the contributing waves and their weights prevents changes in $\widehat{\beta}_k$ from being driven mechanically by changes in reform-wave composition. This aggregation applies the compositional-balance principle of Wing, Freedman, and Hollingsworth (2024) to the wave-specific Sun–Abraham estimates.

The primary estimand is the adoption-boundary mean, estimated as

$$\widehat{\theta}_B = \frac{1}{3} \left(\widehat{\beta}_{-1} + \widehat{\beta}_0 + \widehat{\beta}_{+1} \right). \quad (4)$$

Equation (4) gives equal weight to the three event-time estimates after each has been aggregated across reform waves using Equation (3). I also report the immediate-boundary mean, $(\widehat{\beta}_{-1} + \widehat{\beta}_0)/2$, and the post-adoption mean, $(\widehat{\beta}_0 + \widehat{\beta}_{+1})/2$. Because the primary estimand includes the final national-paper cohort and actual *gaokao* participation is unobserved, it is neither a conventional post-treatment average treatment effect on the treated nor an estimate of the effect of taking a province-set paper. Instead, it summarizes the average change, relative to $k = -2$, in the adopter–comparison gap in later perceived social fairness across the three cohorts nearest the institutional boundary.

Preferred inference uses cluster-robust standard errors with the conventional finite-sample correction (CRV1), clustered by inferred *gaokao*-hukou province, the jurisdictional level of reform assignment, yielding 28 clusters in the preferred sample. Table C.2 of the Supplementary Materials reports alternative CRV1 inference using province-by-expected-examination-year cells as clusters and two-way clustering by inferred *gaokao*-hukou province and expected examination year, together with null-imposed wild-cluster-bootstrap-*t* inference using the same 28 province clusters.

Main Results

Figure 2 presents descriptive profiles of perceived social fairness in adopting provinces for earlier cohorts ($k = -3, -2$) and cohorts nearest the reform boundary ($k = -1, 0, +1$). Each province is compared with a benchmark drawn from nonadopting jurisdictions and reweighted to its expected-examination-year and survey-wave composition. Changes in the adopter–comparison

gap vary across provinces, with pronounced downward shifts in some and smaller or opposite shifts in others.

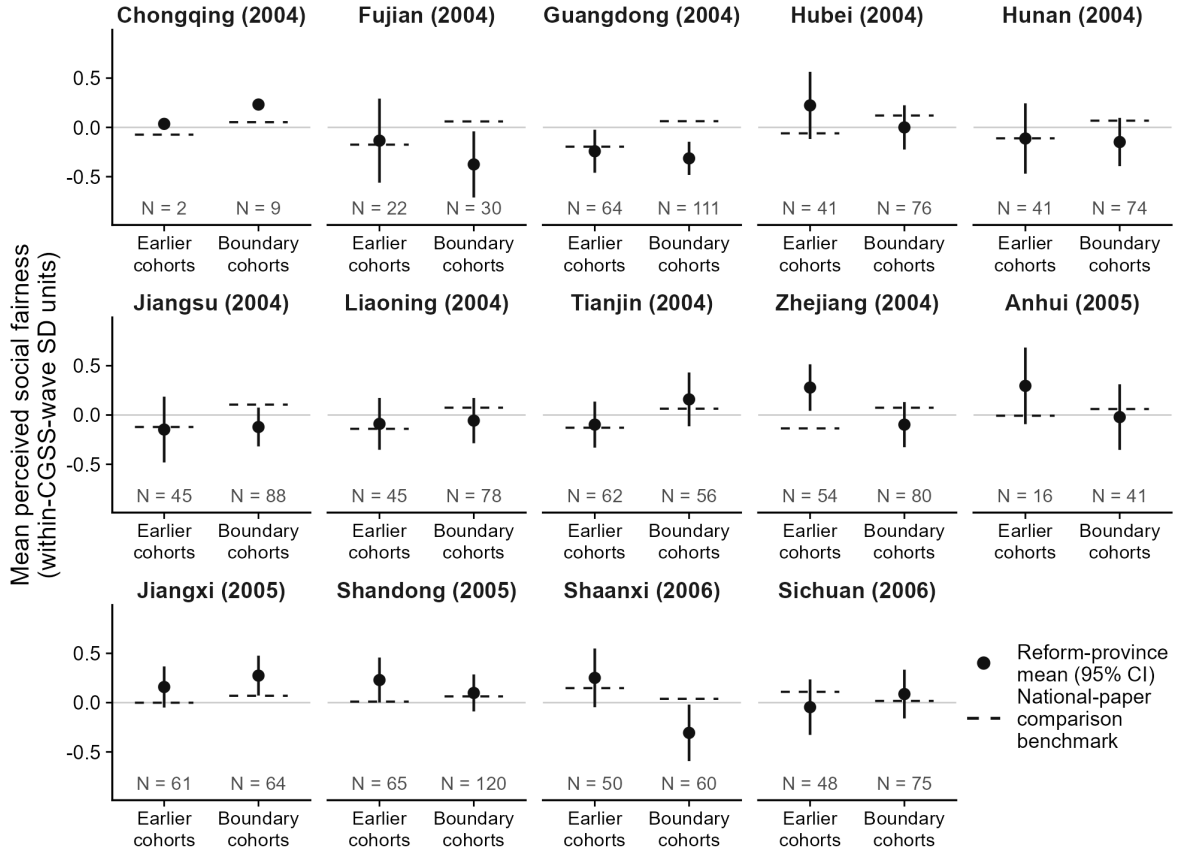


Figure 2: Descriptive Profiles of Perceived Social Fairness around Province-Set-Paper Adoption

Notes: Perceived social fairness is standardized using the mean and standard deviation within the preferred sample in each CGSS wave. Parenthetical years indicate the first province-set-paper examination year. Event time k is expected gaokao year relative to that year; earlier cohorts pool ($k = -3, -2$), and boundary cohorts pool ($k = -1, 0, +1$). Points show unadjusted adopting-province means. Vertical bars show conventional 95% t confidence intervals, omitted when $N < 10$. Dashed segments show national-paper comparison benchmarks. For each adopting province and cohort block, mean standardized perceived social fairness in eligible nonadopting jurisdictions is calculated within expected-examination-year-by-CGSS-wave cells and then averaged using the adopting province's respondent shares across those cells. N is the unweighted number of adopting-province respondents in each cohort block.

Figure 3 presents separate event-study estimates for the preferred sample and the low-education placebo sample. In the preferred sample, the earlier-cohort estimate at $k = -3$ is close to zero, whereas estimates for all three cohorts nearest the reform boundary are negative. The adoption-

boundary mean is $-.212$ points on the original 1–5 scale (95% CI $[-.362, -.062]$, $p = .007$), equivalent to approximately $-.21$ SD. This estimate captures the average change in the adopter–comparison gap in later perceived social fairness across the final national-paper cohort and the first two province-set-paper cohorts.

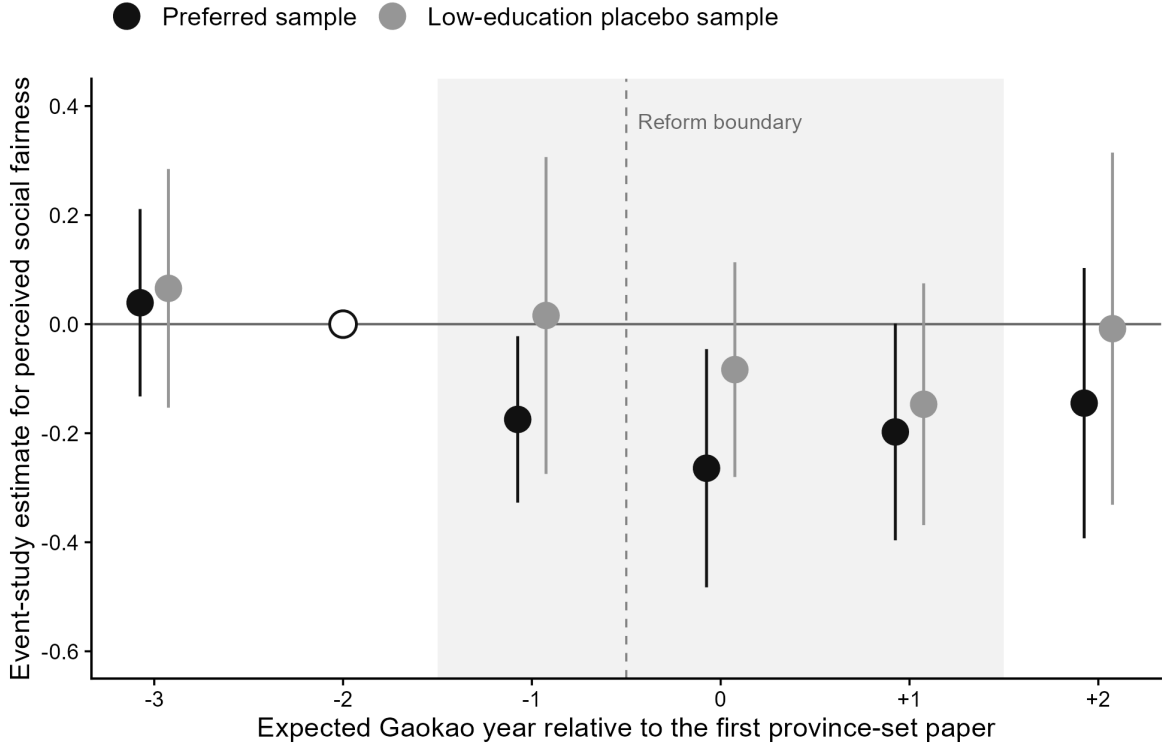


Figure 3: Event-Study Estimates of Perceived Social Fairness around Province-Set-Paper Adoption

Notes: Points report aggregated event-study estimates, $\hat{\beta}_k$, separately for the preferred sample (N=3,387) and the low-education placebo sample (N=2,801). Vertical bars show 95% confidence intervals based on CRV1 standard errors clustered by inferred gaokao-hukou province, with 28 and 29 clusters, respectively. Event time is expected gaokao year relative to the first province-set-paper examination year. The open point at $k = -2$ marks the omitted reference cohort. The dashed vertical line marks the institutional boundary between $k = -1$ and $k = 0$, and shading identifies the $k = -1, 0, +1$ adoption-boundary window. Both samples use the same fixed reform-wave aggregation weights. At $k = +2$, only the 2004 and 2005 adoption waves contribute, with their weights renormalized. Tables C.1–C.2 of the Supplementary Materials report estimates and alternative inference.

The immediate-boundary mean over $k = -1, 0$ is $-.219$ ($SE = .077$), and the post-adoption mean over $k = 0, +1$ is $-.231$ ($SE = .090$); both are statistically distinguishable from zero

(Table C.1 of the Supplementary Materials). The primary adoption-boundary estimate also remains statistically significant under null-imposed wild-cluster-bootstrap- t inference using the same 28 province clusters ($p = .009$). Alternative clustering by province-by-expected-examination-year cells and two-way clustering by inferred *gaokao*-hukou province and expected examination year yield the same conclusion (Table C.2).

In the low-education placebo sample, the adoption-boundary estimate is $-.071$ ($SE = .091$), and neither it nor the immediate-boundary and post-adoption means are statistically distinguishable from zero. Restricting both samples to common province-by-expected-examination-year support leaves the preferred-sample estimate negative and statistically significant, while the placebo estimate remains statistically indistinguishable from zero (Table C.2). The placebo analysis therefore detects no clear adoption-boundary decline among respondents less likely to have prepared for or taken the *gaokao*.

Robustness and Alternative Explanations

The adoption-boundary estimate remains negative and similar in magnitude across alternative assignments of potential reform exposure. This holds when restricting the sample to respondents with observed birth months, assigning birth year plus nineteen when birth month is unavailable, excluding the 2017 and 2018 CGSS waves, retaining only high-confidence hukou assignments, or excluding respondents whose usable sources identify different provinces (Appendix E). Adding gender, Han ethnicity, Chinese Communist Party membership, and income controls likewise leaves the estimate largely unchanged (Appendix F). Every leave-one-province-out estimate also remains negative and close to the preferred estimate (Appendix I).

The estimate is similarly stable when extending the expected-examination-year window to 2000–2008 or allowing a linear cohort trend to vary with provinces' baseline college-enrollment shares, constructed from the 2000 Population Census 1 percent long-form sample (Appendix H). A further check compares extended-window models with and without province-specific linear cohort trends under the same two-cohort normalization. Adding these trends leaves the estimated effect negative and similar in magnitude, but substantially increases the standard error, leaving the

estimate no longer statistically distinguishable from zero.

Timing diagnostics examine whether the pattern aligns with recorded adoption dates. Using a common 1999–2010 cohort sample and fixed specifications and reform-wave weights, I shift adoption dates two or three years earlier or later. None of the four hypothetical-date adoption-boundary means is statistically distinguishable from zero, and the recorded-date estimate is significantly more negative than the nonoverlapping three-years-earlier placebo. In this broader window, the full set of earlier-cohort coefficients at $(k = -7, \dots, -3)$ jointly differs from zero ($p = .029$), whereas the test restricted to $(k = -5, -4, -3)$, where all three adoption waves contribute, does not reject ($p = .527$; Appendix D).

To assess whether the perceived-social-fairness pattern accompanies a broader deterioration in social outlook, I also examine generalized trust and expected exploitation by others. Both show smaller estimated changes, neither statistically distinguishable from zero at the adoption boundary. Pairwise common-sample comparisons also yield larger estimated declines in perceived social fairness, with cross-outcome p -values of .092 for generalized trust and .082 for reverse-coded expected exploitation (Table G.1, Panel B).

A contemporaneous senior-secondary curriculum reform offers a specific alternative explanation because curricular changes can influence students' political attitudes (Cantoni et al. 2017). Its cohort timing, however, differs from the focal paper-setting transitions. The first cohorts to graduate under the new curriculum did so in 2007 in Guangdong, Shandong, Hainan, and Ningxia. In Guangdong and Shandong, the 2007 cohort corresponds to $(k = +3)$ and $(k = +2)$, respectively, outside the adoption-boundary window. Among the focal comparisons, the overlap is confined to Hainan and Ningxia in the 2006-wave comparison at $(k = +1)$ (Appendix A). The two reforms therefore do not overlap in the focal adopting-province cohorts.

Several interpretations of the historical reform remain possible. Centrally prepared papers may be valued as expressions of a common national order as well as shared measures. Territorial standards can render populations legible within a common administrative framework (Scott 1998), and examinations in China have long linked state authority to linguistic, curricular, and political

standardization (Perry 2015; Howlett 2022a). Hierarchical political trust offers another explanation. Citizens commonly report greater confidence in central than in local authorities (Li 2016; Wu and Wilkes 2018), so the decline may partly reflect the transfer of paper-setting authority to the provinces. Changes in perceived examination quality or procedural integrity, disruption to preparation, and coinciding changes in provincial admission opportunities offer further accounts. The following heterogeneity analysis examines whether the perceived-fairness consequences vary with provincial opportunity conditions. Study 2 then varies measurement scope while holding paper-setting authority and formal allocation rules constant.

Exploratory Heterogeneity by Pre-Reform Undergraduate-Institution Density

The shared-measurement account predicts a greater decline in perceived social fairness in less favorable provincial pools. For disadvantaged candidates, withdrawal would remove common evaluative standing and weaken the shared public basis for contesting unequal returns, while leaving their allocation disadvantage unchanged. This expectation is consistent with broader evidence that reactions to decisions can be more sensitive to procedural fairness when outcomes are unfavorable (Brockner and Wiesenfeld 1996).

I examine this implication using pre-reform undergraduate-institution density as a proxy for provincial opportunity conditions. Although universities recruit across provincial boundaries, provincial admission quotas often favor candidates from their home provinces, a pattern documented among elite C9 universities (Hamnett, Shen, and Liang 2019). The geographical distribution of universities therefore matters for opportunities within provincial pools. The measure counts ordinary undergraduate institutions located in each province per 100,000 reported *gaokao* candidates in 2003, capturing local institutional resources relative to candidate demand.

To hold reform timing constant, I rank the nine 2004 adopters and compare the four lowest-density provinces—Hunan, Fujian, Hubei, and Zhejiang—with the four highest-density provinces—Jiangsu, Chongqing, Liaoning, and Tianjin (Table J.2). The joint event-study model retains the full preferred sample, with median-ranked Guangdong and all 2005–2006 adopters represented by nuisance event-time terms.

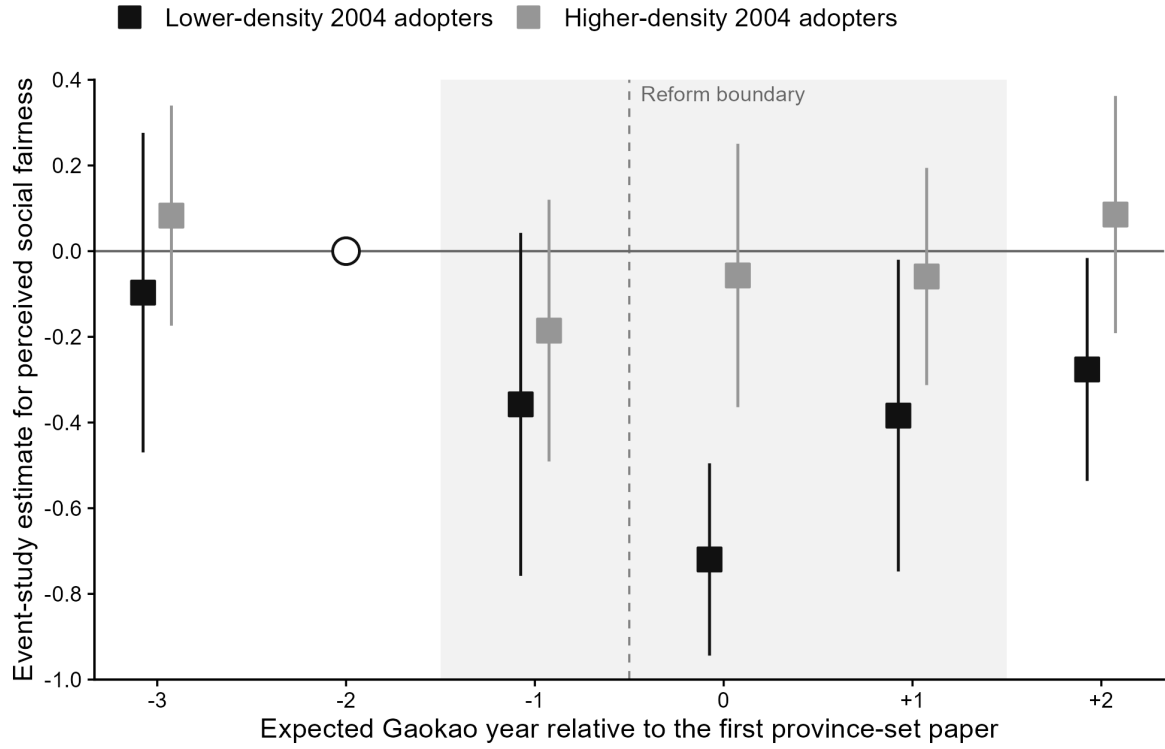


Figure 4: Event-Study Estimates by Pre-Reform Undergraduate-Institution Density among 2004 Adopters

Notes: Points report group-specific event-study estimates from one joint model of the full preferred sample ($N = 3,387$); bars show 95% confidence intervals based on CRV1 standard errors clustered by inferred gaokao-hukou province. Open markers indicate the omitted $k = -2$ reference. The dashed line marks the boundary between $k = -1$ and $k = 0$, and shading identifies the $k = -1, 0, +1$ adoption-boundary window. Appendix J in the Supplementary Materials reports group construction, formal contrasts, and sensitivity checks. Confidence intervals and p -values are unadjusted for the post hoc choice of measure and grouping.

Figure 4 shows a larger adoption-boundary decline in perceived social fairness among lower-density adopters, concentrated in the first province-set-paper cohort. Estimates for higher-density adopters are closer to zero. The formal lower-minus-higher adoption-boundary contrast is $-.387$ points, approximately $-.4$ SD (95% CRV1 CI $[-.580, -.193]$), and is statistically distinguishable from zero under both conventional province-clustered and wild-cluster-bootstrap inference (wild-bootstrap $p = .005$; Table J.1). The earlier-cohort diagnostic at $k = -3$ does not detect a clear group difference. This pattern is consistent with greater fairness costs from withdrawing shared measurement where higher education resources were more limited relative to candidate demand.

Because each focal group contains four provinces, I assess sensitivity to both group assignment and individual provinces. Zhejiang and median-ranked Guangdong have nearly identical densities; replacing Zhejiang with Guangdong in the lower-density group yields a similar negative adoption-boundary contrast (Table J.1). Omitting each of the eight focal provinces in turn, while retaining the original classifications of the remaining provinces, likewise leaves the contrast negative and statistically distinguishable from zero under wild-cluster-bootstrap- t inference (Table J.3).

Study 2: Survey Experiment

Experimental Design and Sample

Study 2 uses a preregistered survey experiment to separate measurement scope from the change in paper-setting authority that accompanied the historical reform.⁹ It also extends the comparison beyond university admission. Respondents evaluate a hypothetical National Young Talent Selection Program that recruits young people for entry-level positions established by central government ministries and centrally administered state-owned enterprises. In both conditions, positions are allocated unequally across provinces, giving applicants in some provinces greater chances of selection. Applicants compete within provincial pools and are selected in descending order of examination scores, with cutoff scores differing across provinces.

Respondents were randomly assigned with equal probability to a nationally uniform-paper condition (A1) or a province-specific-paper condition (A2). In A1, the program's administrative body prepares one examination paper for use nationwide; in A2, the same body prepares a separate paper for each province. In both conditions, this body also organizes, proctors, and grades the examination, and all papers follow a nationally uniform syllabus and have the same maximum score. The vignettes therefore vary the scope of shared measurement while holding paper-setting authority, provincial allocation, and within-province selection rules constant. Appendix L reproduces the

9. The study received an exempt determination from the author's institutional review board, and the pre-analysis plan was registered on OSF before fieldwork. Participation was voluntary and based on informed consent. Appendix L documents privacy protections and the debriefing, which clarified that the talent-selection program was hypothetical.

Chinese-language vignettes and their English translations.

The survey was fielded online in Chinese from August 18 to 20, 2026, among adults residing in mainland China, without demographic quotas. The preregistered pooled completion target was 2,400. Assuming equal allocation and 2,400 nonmissing O1 responses, the preregistered power calculation yielded a minimum detectable effect of approximately .114 standard deviations at 80% power for a two-sided test at $\alpha = .05$ (Appendix K). The provider collected 2,450 completed questionnaires and removed 37 with incorrect responses to the pretreatment reading-confirmation item (AC0) before delivery.¹⁰ All 2,413 delivered records were retained, comprising 1,206 assigned to A1 and 1,207 to A2. Most respondents held a bachelor's degree or higher (87.2%), and nearly all reported having taken the *gaokao* (97.4%; Table M.1). The experiment therefore tests the argument primarily among adults with firsthand experience of high-stakes examination-based selection. Respondents could answer or skip questions and proceed, or exit the survey, but could not return to earlier questions after advancing. Figure 5 summarizes participant disposition and the survey sequence; Appendix K documents recruitment and data delivery.

Measures and Estimation

The sole confirmatory outcome is perceived social fairness (O1), measured immediately after the vignette and before all institution-specific evaluations. Respondents were asked, “Suppose a society primarily uses a system like this to select talent and allocate important opportunities for advancement. Overall, do you think such a society is fair?” Responses range from 1 (“very unfair”) to 5 (“very fair”). The next item measures the perceived fairness of the selection system itself (O2), the preregistered supporting outcome, using the same scale.

Cross-province score comparability (O3) averages two items asking whether applicants' scores can be compared across provinces and used to determine who performed better; both responses must

10. The pre-analysis plan specified termination before randomization for respondents failing AC0; the provider instead applied the screen to completed questionnaires before delivery. AC0 itself preceded assignment. Records and arm-specific counts for randomized respondents who did not complete the questionnaire were unavailable, preventing calculation of completion or attrition rates. In the separate double-list module, every respondent reaching the module encountered the same sensitive item once, with list assignment and order randomized within the A1 and A2 strata.

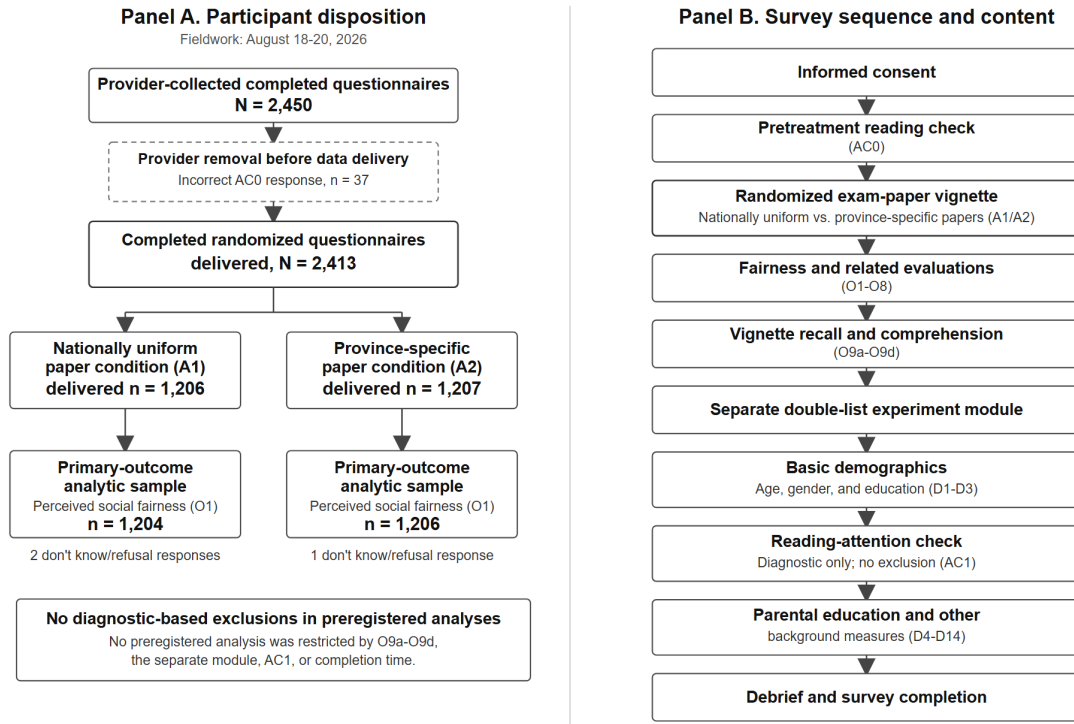


Figure 5: Participant Disposition and Survey Sequence

Notes: Panel A reports the disposition of completed questionnaires. Of 2,450 completed questionnaires collected by the provider, 37 were removed before delivery based on incorrect responses to AC0, leaving 2,413 delivered records. Panel B presents the questionnaire sequence; AC0 was administered before random assignment. The double-list experiment followed O1–O8 and O9a–O9d and belongs to a separate research project with its own pre-analysis plan. Neither its assignments nor its responses enter the analyses reported here. Appendices K–L document recruitment, treatment materials, and measures.

be observed to construct the index. Within-province evaluative accuracy (O4) asks how accurately scores reflect applicants' ability and effort within each province. O3 and O4 are the prespecified nonconfirmatory construct- and discriminant-validity measures, respectively. Preregistered exploratory outcomes assess quota-allocation fairness (O5), opportunities for ordinary people to advance through hard work (O6), rule-violation or cheating risk (O7), and the fairness of the paper arrangement to candidates from different provinces (O8). Higher O7 values indicate greater perceived risk. Table L.1 reports bilingual wording, response scales, coding, and analysis roles.

After the outcome items, O9a assesses recall of the assigned paper arrangement, while O9b–

O9d assess comprehension of features held constant in the vignettes. Treatment is defined exclusively by randomized assignment. No delivered record is excluded from the preregistered outcome analyses based on these diagnostics, subsequent attention checks, list-module responses, or completion time.

Following the pre-analysis plan, I estimate A2–A1 differences in mean ratings within the delivered completed-questionnaire sample using unweighted OLS without covariates. Estimates retain the original 1–5 scales. For the preregistered mean comparisons, I report 95% confidence intervals and two-sided p -values based on HC2 heteroskedasticity-robust standard errors, without adjustment for multiple comparisons. “Don’t know / prefer not to answer” responses and item nonresponse are treated as missing. Each outcome uses its own nonmissing sample without imputation. The O1 analytic sample contains 2,410 respondents; Table M.2 reports outcome-specific missingness by treatment arm.

Main Results

Respondents assigned province-specific papers judge a society relying on the program as less fair than those assigned a common national paper. The A2–A1 difference in perceived social fairness (O1) is $-.151$ points on the five-point scale (95% CI $[-.237, -.066]$, $p < .001$), equivalent to approximately .14 standard deviations using the A1 standard deviation (Glass’s $\delta = -.14$). This finding supports H1 and complements Study 1 by showing a fairness cost of separate papers even when unequal provincial opportunities are explicit and allocation rules and administrative authority remain unchanged in the vignettes. The selection system itself (O2) is also judged less fair, indicating that the response extends to evaluations of both the institution and the society organized around it (Figure 6; Table N.1).

Province-specific papers sharply reduce perceived cross-province score comparability (O3), indicating that respondents regard scores as a weaker common basis for comparing performance across provinces. Respondents also judge the paper arrangement as less fair to candidates from different provinces (O8), with a larger estimated decline than for perceived social fairness. These responses are consistent with the argument that common evaluative terms matter even when candi-

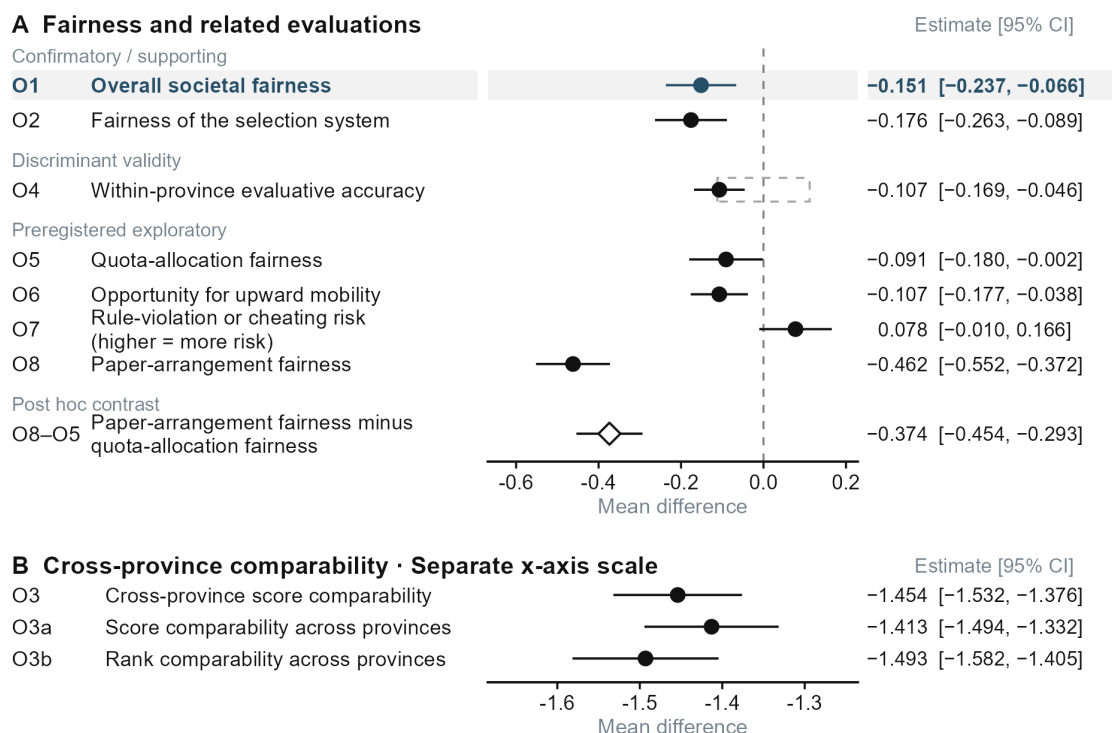


Figure 6: Effects of Examination-Paper Arrangement on Perceived Fairness and Related Evaluations

Notes: Filled points show unweighted OLS mean differences without covariates between the province-specific-paper and nationally uniform-paper conditions (A2 minus A1). Horizontal lines show 95% HC2 confidence intervals; values at right report estimates and intervals. Panel A presents fairness and related evaluations. Panel B presents the construct-validation measure of cross-province comparability (O3) and its component items (O3a and O3b) on a separate x-axis scale. Outcome ratings retain their original 1–5 scales; O3 averages O3a and O3b, and higher O7 values indicate greater perceived risk. Shading highlights O1, the sole confirmatory outcome. The dashed box for O4 marks the prespecified equivalence bounds of ± 0.15 times its A1 standard deviation; the equivalence test did not establish equivalence. The open diamond shows the post hoc treatment-arm difference in respondent-level O8–O5 scores among complete pairs. Outcome-specific analytic samples range from 2,347 to 2,413 respondents; the O8–O5 contrast uses 2,391 complete pairs. Table N.1 reports the preregistered estimates, and Table P.2 reports the paired contrast.

dates compete in separate allocation pools.

The response also extends to perceptions of within-province evaluative accuracy (O4). Respondents assigned province-specific papers express less confidence that scores reflect applicants' ability and effort within each province. Although much smaller than the decline in cross-province

comparability, this response departs from the anticipated discriminant-validity pattern, and the pre-specified test does not establish equivalence.¹¹ This decline may reflect concerns about the accuracy of province-specific assessment, spillover from broader judgments of procedural fairness, or both.

The exploratory outcomes also show how the response extends to broader judgments of opportunity. Respondents perceive fewer opportunities for ordinary people to advance through hard work (O6). Quota-allocation fairness (O5) also declines, although its confidence interval only narrowly excludes zero, providing more tentative evidence of spillover to an unchanged distributive rule. A post hoc comparison using respondents' paired O8 and O5 ratings shows that paper differentiation produces a larger decline in paper-arrangement fairness than in quota-allocation fairness (Table P.2). The fairness response is thus concentrated more strongly on how candidates are evaluated than on the unchanged allocation of opportunities. Evidence of increased concern about misconduct is less clear. The estimated effect on rule-violation or cheating risk (O7) is positive but not statistically distinguishable from zero at the .05 level ($p = .084$).

Robustness and Diagnostics

Post hoc analyses examine whether the O1 finding depends on demographic adjustment, model choice, or item nonresponse. Adjustment for categorical age group, gender, and education leaves the estimate essentially unchanged relative to the unadjusted model fitted to the same complete-case sample (Table O.1). A proportional-odds ordered-logit model likewise indicates lower perceived social fairness under province-specific papers (Table O.2). Extreme-value assignments to the three missing O1 responses cannot reverse the negative mean contrast. O7 is less conclusive: don't-know/refusal responses are more frequent in A1, and the corresponding bounds span zero. These bounds concern item nonresponse within delivered questionnaires; they do not address unobserved survey attrition (Table M.2).

Because respondents may draw on their own examination experiences when evaluating the hy-

11. The two one-sided tests use bounds of ± 0.15 times the observed O4 standard deviation among A1 respondents in the O4 analytic sample. The 90% HC2 confidence interval is not wholly contained within these bounds. Equivalence is therefore inconclusive; this does not establish a difference exceeding the prespecified margin (Table N.1, Panel D).

pothetical program, I also examine the O1 contrast among prior examination participants. Separate post hoc restrictions to respondents reporting *gaokao* or public-sector examination participation retain the negative contrast. Preregistered exploratory interactions detect no clear differences by either experience, although the small number of *gaokao* nonparticipants makes that comparison particularly imprecise (Table O.1). Additional descriptive checks document treatment-arm composition, geographic coverage, completion times, and the positive within-arm association between the two O3 items (Table M.1; Figures M.1 and P.1; Table P.2, Panel B).

The factual diagnostics distinguish recognition of the assigned paper arrangement from comprehension of unchanged features. Recall of the paper arrangement (O9a), provincial quota allocation (O9c), and within-province score ranking (O9d) is high in both arms, with no detectable between-arm differences. Pass rates on the subsequent reading-attention check (AC1) are also high. Correct recall of the common syllabus and maximum score (O9b), however, falls from 90.5% in A1 to 70.5% in A2. The largest shift toward an incorrect response is toward different syllabi with the same maximum score (Table P.1). Respondents generally recognize the manipulated arrangement and unchanged allocation rules, but do not retain all features of the common examination framework equally across conditions.

The O9b discrepancy may reflect limited attention to or recall of the common-syllabus statement, although the diagnostic does not establish its source. Descriptive post hoc comparisons within each of the four O9-correct subsets retain negative O1 contrasts, including among respondents correctly recalling both the common syllabus and maximum score.¹² These comparisons are reported in Table P.2, Panel C.

12. These analyses condition on post-treatment responses and do not preserve randomized comparability. They were added after inspection of the formal results and depart from the pre-analysis plan's restriction on such conditioning. Table P.3 documents this deviation; the preregistered O1 analysis remains unchanged.

Conclusion

High-stakes selection institutions determine not only how opportunities are distributed but also how far common evaluative terms extend across allocation boundaries. This paper shows that this second dimension matters to perceived fairness. Study 1 links China's mid-2000s transition to province-set *gaokao* papers to lower perceived social fairness measured years later among cohorts nearest the reform boundary. Study 2 finds a corresponding fairness cost of province-specific papers in a pre-registered experiment involving a hypothetical national talent-selection program, with allocation, within-province selection rules, and paper-setting authority held constant in the vignettes. These studies support an *evaluative fairness premium*: an unequal opportunity order can be judged fairer when candidates from separate allocation pools stand before a common measure.

Research on responses to inequality emphasizes perceived distributions and the uneven effects of information on concern about inequality and redistributive preferences (Gimpelson and Treisman 2018; Kuziemko et al. 2015). In Study 2, unequal provincial quotas, unequal selection chances, and different cutoff scores were explicitly described in the same terms across conditions. Fairness judgments nevertheless differed with the scope of shared measurement. This pattern is consistent with relational accounts of procedural justice, in which treatment by authorities conveys people's standing within a group (Tyler and Lind 1992). The experiment does not, however, identify the psychological pathways underlying this effect; perceived within-province evaluative accuracy may also contribute to the fairness response. The institutional contribution is not simply that procedures matter alongside distributions. It is that common evaluative treatment can matter across the boundaries separating candidates' access to opportunities, even when those candidates do not compete directly with one another.

The findings also speak to how citizens interpret state action. Lü (2014) distinguishes policy awareness from material benefits, showing how media reporting and the attribution of policy credit help shape evaluations of government. His account treats citizens as interpreters of state action, not simply recipients of its benefits. Building on this perspective, the present paper identifies the

organization of meritocratic selection as a basis for public judgments of the wider opportunity order. Those judgments can turn not only on the benefits the state provides and who receives credit for them, but also on whether its institutions assess candidates across unequal allocation pools through a common measure.

The political stakes of institutional design under authoritarian rule can extend beyond the moment of reform. Research on repression documents attitudinal consequences that endure for decades and can extend beyond the generation directly exposed to violence (Desposato, Wang, and Wu 2021; Wang 2021). Related work shows that contemporary conflicts can reactivate collective memories and make them consequential for behavior (Fouka and Voth 2023). Study 1 brings the question of longer-term attitudinal consequences to a nonviolent but consequential institutional encounter. The transition to province-set *gaokao* papers is associated with lower perceived social fairness measured years later among cohorts nearest the reform boundary. Such fairness judgments bear on the psychological foundations of legitimacy, since beliefs that existing social arrangements are appropriate and just can sustain acceptance of an unequal order (Tyler 2006).

China's 2014 decision to expand the use of nationally written *gaokao* papers offers a further opportunity to examine these implications. The subsequent expansion extended centrally written paper variants across provincial boundaries while admissions remained province-based. Officials emphasized uneven paper quality and the costs of decentralized paper-setting, but the present findings identify an additional public stake in this design choice. The evidence does not establish how that later expansion affected perceived social fairness. Future research could examine whether renewed shared measurement improved fairness judgments among later cohorts and whether it also changed the evaluations of those who had experienced its earlier withdrawal.

Although this paper emphasizes the political stakes of institutional design under authoritarian rule, the distinction between allocation and measurement also has implications for other settings. Large-scale meritocratic selection structures access to public employment in Indonesia through civil-service examinations (Kuipers 2023). Research on Indian engineering and medical admissions and reservation systems further shows how common examinations and merit rankings can coexist

with group-specific eligibility and allocation rules (Bagde, Epple, and Taylor [2016](#); Jeffery [2024](#); Sönmez and Yenmez [2022](#)). The findings here direct attention to the public meaning of shared evaluative standing alongside the distributive purposes and consequences of such arrangements. Understanding perceived fairness therefore requires asking not only who gets what, but also who stands before the same measure.

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SUPPLEMENTARY MATERIALS

for

*Equal Before the Measure:
Shared Measurement and Perceived Fairness
in an Unequal Order*

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Appendix A: Province-Set Gaokao Papers: Adoption Timing and Analysis Groups

Table A.1. Province-Set-Paper Adoption and Comparison Provinces

Adopting provinces	First province-set-paper examination year	Comparison provinces
Chongqing	2004	Gansu
Fujian	2004	Guangxi
Guangdong	2004	Guizhou
Hubei	2004	Hainan
Hunan	2004	Hebei
Jiangsu	2004	Heilongjiang
Liaoning	2004	Henan
Tianjin	2004	Inner Mongolia
Zhejiang	2004	Jilin
Anhui	2005	Ningxia
Jiangxi	2005	Qinghai
Shandong	2005	Shanxi
Shaanxi	2006	Tibet
Sichuan	2006	Xinjiang
		Yunnan

Notes: The table classifies the 29 province-level jurisdictions eligible for Study 1 after excluding Beijing and Shanghai at the policy level and before respondent-level restrictions. For each adoption wave, the adopting group consists of jurisdictions first adopting in that year, while eligible nonadopting jurisdictions form the comparison group. Tibet contributes no observations to the preferred sample, leaving 28 province clusters, but appears in the low-education placebo sample.

Public-notice timing differed from implementation timing. In all adopting jurisdictions except Sichuan, the earliest public notice I have located postdated the final national-paper cohort’s examination. Sichuan is the exception because a January 2005 notice preceded the 2005 examination. “Earliest public notice located” refers to the earliest public-facing notice identified in the source search and does not rule out earlier internal communication.

The first curriculum-reformed Gaokao cohorts graduated in 2007 in Guangdong, Shandong, Hainan, and Ningxia. This timing does not overlap the $k = -1, 0, +1$ focal treated cohorts in Guangdong, which adopted province-set papers in 2004, or Shandong, which adopted in 2005: the 2007 cohort corresponds to $k = +3$ and $k = +2$, respectively. The overlap with the adoption-boundary estimand is confined to Hainan and Ningxia, which enter the 2006-wave comparison group at $k = +1$. The curriculum reform therefore does not coincide with the focal treated cohorts and overlaps only part of one comparison cell.

Appendix B: Sample Construction

This appendix documents the construction of the two variables used to link respondents to province-by-cohort examination-paper arrangements: expected *gaokao* year and inferred *gaokao*-hukou province. Study 1 draws on eight waves of the Chinese General Social Survey (CGSS), fielded in 2010, 2011, 2012, 2013, 2015, 2017, 2018, and 2021. The inferred *gaokao*-hukou province approximates the household-registration jurisdiction most likely to have governed the respondent near examination age. It is distinct from the province in which the respondent was surveyed, which is retained separately for the survey-province-by-CGSS-wave fixed effects.

The CGSS does not record whether a respondent actually took the *gaokao*, the year in which the respondent took it, or the province in which the respondent registered for the examination. Nor does it measure awareness or recall of the province-set-paper reform. The resulting province-by-cohort assignment therefore captures potential exposure associated with an inferred province and expected examination year, not observed examination participation.

Because the available household-registration and migration questions differ across CGSS waves, I use a fixed hierarchy to infer the province most likely to have been relevant near examination age. I first construct a candidate for current hukou province. A directly reported province is used when available. When it is unavailable, survey province substitutes only if the respondent reports that the hukou is registered in the surveyed locality. Current residence alone is therefore not treated as evidence of current hukou. In the 2013, 2015, and 2018 waves, city- or county-level information is also used to recover province, but only when the geographic mapping is unambiguous.

The algorithm then retains the first valid province-level unit in the following order: S1, current hukou province for respondents whose hukou has remained in the current locality since birth; S2, current hukou province for respondents whose hukou was transferred to the

current locality before age 16; S3, hukou province at age 14; S4, hukou province before moving to the current locality; S5, hukou province at birth; S6, usual-residence province at age 14; S7, usual-residence province before moving to the current locality; and S8, the mother’s primary province of residence at the respondent’s birth.

S1–S5 rely on hukou information, whereas S6–S8 are residence-based proxies used only when higher-priority hukou information is unavailable. S1–S3 form the high-confidence tier used in the corresponding robustness analysis. Because transfer timing is reported only by calendar year, an implied transfer age of exactly 16 does not establish that the move occurred before age 16 and therefore does not satisfy S2. Such respondents continue through the remaining rules supported by the relevant survey wave.

Once a respondent is assigned under one rule, lower-priority sources do not alter the assignment. Respondents are not excluded when two or more usable sources identify different provinces. Instead, the highest-priority valid source determines the assigned province, and a disagreement indicator records the case for diagnostic and sensitivity analyses. Respondents for whom no rule produces a valid province remain in the cleaned data with assignment code S0. Table B.1 reports incremental assignments under this hierarchy: entries for S1–S8 indicate the number of respondents first assigned by each rule, not the total number for whom the corresponding item is observed.

The available information varies substantially across waves. The 2010, 2011, 2017, and 2021 waves support only S1–S2, principally through the locally registered hukou fallback; their province-identification rates range from 66.9% to 71.3%. The 2012 and 2013 waves additionally support S7–S8; the 2013 wave also permits unambiguous geographic repairs, and both waves identify 99.8% of respondents. The 2015 wave supports S1–S2 and S7, supplemented by unambiguous city- or county-to-province repairs, and identifies 96.6%. The 2018 wave supports the complete S1–S8 hierarchy through current-province, age-14, pre-move, birth, maternal, and unambiguous geographic information, identifying 99.9% of respondents.

Across all eight waves, this procedure assigns an inferred *gaokao*-hukou province to 73,136

of 85,091 respondents, or 86.0%. The remaining 11,955 respondents are coded S0. Among the 73,136 assigned respondents, 1,991 have at least two usable sources that identify different provinces; the hierarchy retains the higher-priority assignment. This group includes 111 respondents in the preferred estimation sample.

Expected *gaokao* year is assigned from year and month of birth. Respondents born from January through August are assigned birth year plus 18. Respondents born from September through December are assigned birth year plus 19. When year of birth is observed but month of birth is missing, the point assignment is birth year plus 18. I also retain birth year plus 18 and birth year plus 19 as lower and upper assignments for sensitivity analysis. The public-use files for the 2017 and 2018 CGSS waves provide no birth-month information. Every respondent with a valid birth year in those waves therefore receives the plus-18 point assignment and the same plus-18 and plus-19 bounds. Five respondents lack a valid birth year and cannot be assigned an expected examination year. Table B.3 reports the exact assignment counts by wave.

Expected examination year is a cohort-based approximation rather than an observed examination date. It cannot capture early or delayed school entry, repeating a school grade, spending one or more additional years preparing to retake the *gaokao*, or otherwise taking the examination at an atypical age. Appendix E evaluates sensitivity to these assignments by restricting the sample to respondents with an observed birth month, assigning plus 19 rather than plus 18 when month is unavailable, and excluding the 2017 and 2018 waves.

The eight CGSS waves contain 85,091 respondents. Expected *gaokao* year can be assigned to 85,086, and an inferred *gaokao*-hukou province can be assigned to 73,136. Five respondents lack a valid birth year, 11,955 lack an inferred province, and three fall into both categories. Requiring expected year, inferred *gaokao*-hukou province, and survey province to be jointly observed therefore leaves 73,134 respondents. This distinction explains why the number with an inferred province, 73,136, is slightly larger than the jointly observed sample.

I next restrict the data to eligible province-by-year policy cells, expected *gaokao* years

from 2000 through 2008, and provinces other than Beijing and Shanghai. Beijing and Shanghai are excluded because they adopted province-set papers before the 2004–2006 reform waves examined in Study 1. This extended-window sample contains 8,125 respondents, all of whom match a province-by-year cell in the examination-paper policy file. Restricting expected *gaokao* year to the preferred 2001–2007 window leaves 6,207 respondents.

I define the potential-exposure education group as respondents whose reported highest educational level is secondary specialized school, vocational high school, technical school, general senior high school, or any higher level. This definition identifies respondents for whom personal exposure to the *gaokao* was plausible; it does not establish that every respondent sat the general examination. The restriction yields 3,393 respondents. Six lack the perceived-social-fairness outcome, producing the preferred estimation sample of 3,387 respondents.

Before the outcome restriction, a mutually exclusive group of 2,811 respondents with junior high school education or less forms the low-education category. Ten lack perceived social fairness, leaving 2,801 respondents in the low-education placebo sample. These respondents had weaker reasons for direct personal exposure to the examination, but they are not assumed to have been wholly unaware of the reform. The group is therefore used as a population-placebo sample, not as the cross-jurisdictional comparison group or a strictly untreated population. Three additional respondents have missing or nonresponse education items and cannot be classified into either education group. Together with the six outcome-missing respondents in the potential-exposure group and the ten in the low-education group, these counts exhaust the 6,207 respondents in the preferred cohort window. The preferred estimation sample covers 28 inferred *gaokao*-hukou provinces, whereas the low-education placebo sample covers 29; the difference arises because Tibet contributes observations only to the placebo sample.

Table B.1. Ordered Rules for Inferring Gaokao-Hukou Province and Incremental Assignments, by CGSS Wave

Rule	Information used	Evidence tier	2010	2011	2012	2013	2015	2017	2018	2021	All waves	Preferred sample
S1	Current hukou province; hukou remained in current locality since birth	High confidence	8,212	3,602	8,078	8,353	7,426	8,507	8,514	5,557	58,249	2,266
S2	Current hukou province; transferred to current locality before age 16	High confidence	195	156	222	175	239	227	331	146	1,691	85
S3	Hukou province at age 14	High confidence	—	—	—	—	—	—	3,911	—	3,911	327
S4	Hukou province before moving to current locality	Direct hukou	—	—	—	—	—	—	12	—	12	1
S5	Hukou province at birth	Direct hukou	—	—	—	—	—	—	0	—	0	0
S6	Usual-residence province at age 14	Own-residence proxy	—	—	—	—	—	—	4	—	4	0
S7	Usual-residence province before moving	Own-residence proxy	—	—	3,136	2,541	2,932	—	1	—	8,610	661
S8	Mother's primary residence province at respondent's birth	Maternal-residence proxy	—	—	310	348	—	—	1	—	659	47
Total identified			8,407	3,758	11,746	11,417	10,597	8,734	12,774	5,703	73,136	3,387
Candidate-source disagreement			—	—	548	556	73	—	814	—	1,991	111
Still unresolved (S0)			3,376	1,862	19	21	371	3,848	13	2,445	11,955	—
Respondents in displayed universe			11,783	5,620	11,765	11,438	10,968	12,582	12,787	8,148	85,091	3,387
Identification rate			71.3%	66.9%	99.8%	99.8%	96.6%	69.4%	99.9%	70.0%	86.0%	—

Notes: Entries for S1–S8 are incremental assignments under the hierarchy described in the text. An em dash indicates that no usable candidate from that information source is observed in a wave; 0 indicates that the source is observed but adds no respondent after higher-priority rules. Candidate-source disagreement counts identified respondents for whom at least two usable sources name different provinces; the higher-priority source determines the assignment. The Preferred sample column reports rule assignments among the 3,387 respondents in the main estimation sample; dashes in its diagnostic rows denote not applicable.

Table B.2. Construction of the Preferred and Low-Education Placebo Samples, by CGSS Wave

Construction step	2010	2011	2012	2013	2015	2017	2018	2021	Total
Raw CGSS respondents	11,783	5,620	11,765	11,438	10,968	12,582	12,787	8,148	85,091
Expected Gaokao year assigned	11,780	5,620	11,764	11,437	10,968	12,582	12,787	8,148	85,086
Inferred Gaokao-hukou province assigned	8,407	3,758	11,746	11,417	10,597	8,734	12,774	5,703	73,136
Survey province identified	11,783	5,620	11,765	11,438	10,968	12,582	12,787	8,148	85,091
Expected year, inferred Gaokao hukou, and survey province jointly available: jointly observed sample	8,406	3,758	11,745	11,417	10,597	8,734	12,774	5,703	73,134
Location eligible and expected Gaokao year in 2000–2008	857	374	1,334	1,394	1,207	856	1,465	638	8,125
Province-by-year exam-paper policy cell matched	857	374	1,334	1,394	1,207	856	1,465	638	8,125
Expected Gaokao year in 2001–2007	667	289	1,014	1,059	895	652	1,122	509	6,207
Education consistent with potential Gaokao exposure	348	140	595	586	497	325	643	259	3,393
Perceived social fairness nonmissing: preferred estimation sample	348	139	595	585	496	323	642	259	3,387
Low-education placebo sample with perceived social fairness nonmissing	317	149	419	473	393	325	476	249	2,801

Notes: The first three identification rows are separate coverage diagnostics; their intersection is the jointly observed sample. From that row through the preferred estimation sample, the counts form a monotone funnel. Extended-window eligibility requires an expected Gaokao year in 2000–2008 and excludes Beijing and Shanghai. The main window is 2001–2007. The potential-exposure education group comprises respondents whose highest education is secondary specialized school, vocational high school, technical school, general senior high school, or higher. The low-education placebo sample comprises respondents with junior high school or less and nonmissing perceived social fairness.

Table B.3. Expected Gaokao-Year Assignment, by CGSS Wave

Wave	Jan.–Aug. birth year +18	Sept.–Dec. birth year +19	Individual month missing +18	Wave has no birth-month information +18	Birth year missing	Total
2010	7,660	4,107	13	0	3	11,783
2011	3,562	1,981	77	0	0	5,620
2012	7,484	4,113	167	0	1	11,765
2013	7,447	3,751	239	0	1	11,438
2015	6,719	3,523	726	0	0	10,968
2017	0	0	0	12,582	0	12,582
2018	0	0	0	12,787	0	12,787
2021	5,095	2,643	410	0	0	8,148
Total	37,967	20,118	1,632	25,369	5	85,091

Notes: January–August births are assigned birth year +18 and September–December births birth year +19. Missing individual birth month uses +18 as the point assignment; +18 and +19 are retained as the lower and upper expected-year bounds. CGSS 2017 and 2018 do not provide birth month, so valid birth years in those waves also use +18 as the point assignment and retain the same +18/+19 bounds. Rule counts add exactly to the raw count in every wave.

Appendix C: Event-Study Results and Cluster-Robust Inference

Table C.1. Event-Study Estimates: Preferred and Low-Education Samples

DV	Perceived Social Fairness (1–5; higher values indicate greater perceived fairness)			
	Preferred sample: potential Gaokao exposure		Low-education placebo sample	
Model				
Event time	Estimate (SE)	Adopting- province N	Estimate (SE)	Adopting- province N
–3	0.039 (0.084)	322	0.066 (0.107)	255
–2	(<i>reference</i>)	294	(<i>reference</i>)	229
–1	–0.175* (0.074)	311	0.016 (0.142)	209
0	–0.264* (0.106)	301	–0.083 (0.096)	207
+1	–0.198† (0.097)	350	–0.147 (0.108)	189
+2	–0.145 (0.121)	328	–0.008 (0.158)	121
Adoption-boundary mean: $k = -1, 0, +1$	–0.212** (0.073)	962	–0.071 (0.091)	605
Immediate-boundary mean: $k = -1, 0$	–0.219** (0.077)	612	–0.034 (0.099)	416
Post-adoption mean: $k = 0, +1$	–0.231* (0.090)	651	–0.115 (0.096)	396
Earlier-cohort diagnostic: $H_0 : \beta_{-3} = 0$	$p = .643$	322	$p = .544$	255
Individual controls	No		No	
Expected Gaokao-year window	2001–2007		2001–2007	
Observations	3,387		2,801	
Adopting-province N	2,260		1,463	
Province clusters	28		29	
Inferred Gaokao-hukou province FE	Yes		Yes	
Expected Gaokao year FE	Yes		Yes	
Survey province $\times$ CGSS wave FE	Yes		Yes	

Notes: Event time k is expected Gaokao year minus the relevant reform wave’s adoption year, with $k = -2$ omitted. Neither sample uses CGSS sampling weights. Both use the same fixed reform-wave aggregation weights, based on the preferred sample’s adopting-province respondent shares at $k = -2$ (2004: .599; 2005: .228; 2006: .173). All three waves contribute from $k = -3$ through $+1$; at $k = +2$, weights are renormalized over the 2004 and 2005 waves. Reported means equally weight the event-time coefficients specified in the row labels. Adopting-province N counts unweighted respondents in the 14 adopting provinces; N for each mean sums the corresponding event-cell counts. Education groups are defined in Table B.2. FE denotes fixed effects. CRV1 standard errors, in parentheses, are clustered by inferred Gaokao-hukou province. † $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Table C.2. Cluster-Robust Inference for the Adoption-Boundary Estimand

DV	Perceived Social Fairness (1–5; higher values indicate greater perceived fairness)					
Sample / inference	Adopting- province N $k = -1, 0, +1$	Estimate	SE	95% CI	p	Clusters / df
<i>Panel A. Preferred Province-Level Inference</i>						
Province clusters (CRV1)	962	-0.212	0.073	[-0.362, -0.062]	.007	$G = 28$ $df = 27$
Wild-cluster-bootstrap- t	962	-0.212	–	[-0.374, -0.063]	.009	$G = 28$ $B = 9, 999$
<i>Panel B. Alternative Clustering Schemes</i>						
Province $\times$ expected-Gaokao-year cell clusters (CRV1; secondary)	962	-0.212	0.071	[-0.352, -0.072]	.003	$G_{p \times y} = 191$ $df = 190$
Two-way clusters: inferred province and expected Gaokao year (CRV1; secondary)	962	-0.212	0.040	[-0.310, -0.114]	.002	$G_p = 28$ $G_y = 7$ $df = 6$
<i>Panel C. Common-Support Preferred and Low-Education Placebo Samples</i>						
Preferred sample (CRV1)	947	-0.222	0.069	[-0.363, -0.080]	.003	$G = 28$ $df = 27$
Low-education placebo sample (CRV1)	605	-0.074	0.091	[-0.261, 0.113]	.425	$G = 28$ $df = 27$

Notes: All estimates use the adoption-boundary estimand, reform-wave weights, and fixed-effects specification defined in Table C.1. Adopting-province N counts unweighted respondents at $k = -1, 0, +1$. Panel A reports CRV1 and null-imposed wild-cluster-bootstrap- t inference clustered by inferred *gaokao*-hukou province; the bootstrap uses Rademacher weights and 9,999 replications. Panel B changes only the covariance estimator. With only seven expected-year clusters, the two-way result is treated as a sensitivity diagnostic; its covariance matrix incorporates *fixest*'s eigenvalue repair. Panel C separately re-estimates the preferred and low-education placebo samples on the same 186 province-by-expected-year cells across 28 provinces, using province-clustered CRV1 inference.

Appendix D: Hypothetical Adoption-Date and Temporal-Localization Tests

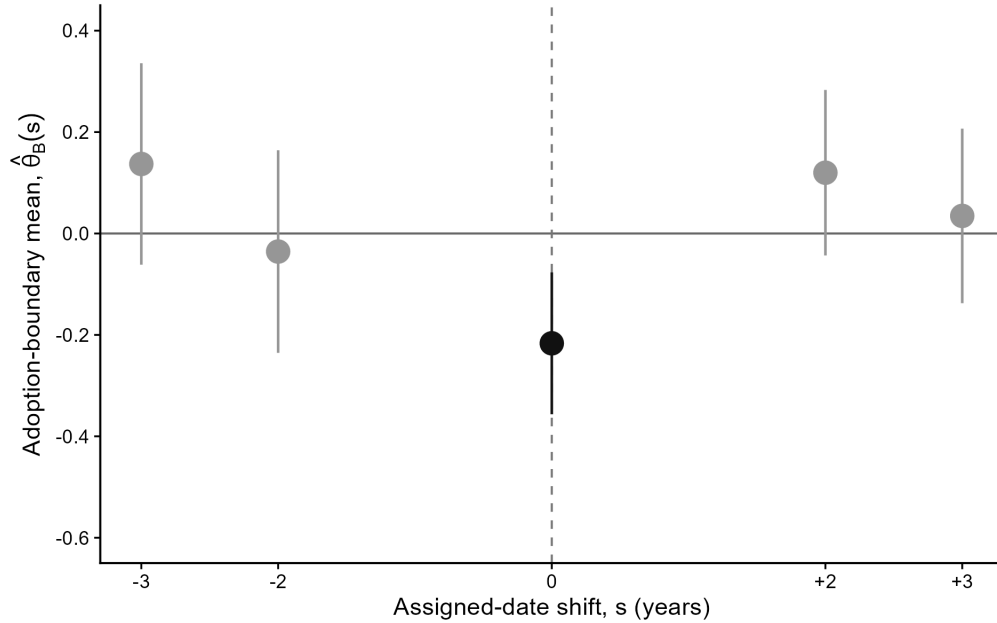
To assess whether the adoption-boundary contrast is localized around the recorded boundary, I shift each adopting province’s adoption year two or three years earlier or later and re-estimate the same Sun–Abraham specification on an expanded 1999–2010 expected-Gaokao-year window. The preferred education and outcome-availability restrictions yield 5,888 respondents across 28 inferred Gaokao-hukou provinces. Across five refits, the sample, fixed effects, clustering, and Table C.1 reform-wave weights remain fixed; only assigned dates and pseudo-event indicators change. All waves support the $k^* = -2$ reference and $k^* = -1, 0, +1$ boundary. The $s = -3$ window is a non-overlapping pre-implementation timing placebo; $s = -2$ is a near-boundary diagnostic that includes recorded $k = -1$. Later shifts compare post-adoption cohorts, not whether the effect has disappeared. Because fairness was measured later, this varies cohort-based exposure rather than pretreatment attitudes. Figure D.1 compares the boundary means; Tables D.1 and D.2 report the event paths and map pseudo-event time into recorded event time.

The recorded-date estimate is -0.217 ($SE = 0.068$; 95% CI $[-0.356, -0.077]$; CRV1 $p = .004$; wild-cluster-bootstrap- t $p = .008$). The $s = -3$ estimate is 0.137 ($SE = 0.097$), and none of the four hypothetical-date means is distinguishable from zero at the .05 level under either inference method. The recorded-minus- $s = -3$ difference, $\hat{\theta}_B(0) - \hat{\theta}_B(-3)$, is -0.354 ($SE = 0.156$; 95% CI $[-0.675, -0.033]$; CRV1 $p = .032$; wild-cluster-bootstrap- t $p = .049$). The five refit estimates equal the corresponding loadings on the recorded-date saturated event path, so I compute this difference as a linear restriction using its full province-clustered covariance matrix; the bootstrap imposes $H_0 : \theta_B(0) - \theta_B(-3) = 0$.

In the same recorded-date model and sample, a joint CRV1 test of $k = -7, \dots, -3$ gives $F(5, 27) = 2.980$ and $p = .029$. The $k = -7$ coefficient uses only the 2006 wave and

$k = -6$ the 2005 and 2006 waves, mapping to expected Gaokao years 1999–2000 outside the preferred window. Only $k = -6$ differs from zero at the nominal .05 level in a pointwise two-sided CRV1 test without adjustment for multiple testing (estimate = -0.189 ; 95% CI $[-0.366, -0.011]$; $p = .038$), which does not establish that it alone accounts for the joint rejection. The common-support coefficients at $k = -5, -4, -3$ are $-0.136, -0.018$, and 0.021 . Although the full set of earlier-cohort coefficients jointly differs from zero, the joint test does not reject equality to zero among the pre-boundary cohorts with common support across all three adoption waves ($F(3, 27) = 0.759$, $p = .527$). The latter result is consistent with the local parallel-cohort-trends assumption.

Figure D.1. Adoption-Boundary Estimates under Recorded and Hypothetical Adoption Dates



Notes: Points report $\hat{\theta}_B(s)$, the equally weighted mean of the pseudo-event-time coefficients at $k^* = -1, 0, +1$, relative to $k^* = -2$, after shifting each adopting province's recorded adoption date by s years and re-estimating the same Sun–Abraham specification with the respondent sample, fixed-effects structure, and reform-wave aggregation weights held fixed. Bars show 95% confidence intervals based on CRV1 standard errors clustered by inferred Gaokao-hukou province. The black point at $s = 0$ uses the recorded adoption dates; gray points use hypothetical dates. The horizontal line marks zero, and the vertical dashed line marks $s = 0$, the recorded-date specification.

Table D.1. Event-Study Estimates under Hypothetical Adoption Dates

DV	Perceived Social Fairness (1–5; higher values indicate greater perceived fairness)			
	Three years early ($s = -3$)	Two years early ($s = -2$)	Two years late ($s = +2$)	Three years late ($s = +3$)
Assigned-date shift	Estimate (SE)	Estimate (SE)	Estimate (SE)	Estimate (SE)
Pseudo event time, k^*				
–3	–0.171 [†] (0.099)	–0.118 (0.084)	0.104 (0.095)	–0.108 (0.089)
–2	(reference)	(reference)	(reference)	(reference)
–1	0.118 (0.084)	0.040 (0.104)	0.108 (0.089)	0.055 (0.084)
0	0.158 (0.126)	0.018 (0.110)	0.162 [†] (0.091)	–0.018 (0.093)
+1	0.136 (0.121)	–0.165 (0.109)	0.090 (0.098)	0.067 (0.106)
+2	–0.047 (0.102)	–0.269* (0.109)	0.175 (0.104)	–0.028 (0.137)
Adoption-boundary mean: $k^* = -1, 0, +1$	0.137 (0.097)	–0.036 (0.097)	0.120 (0.080)	0.035 (0.084)
95% CI	[–0.062, 0.336]	[–0.235, 0.164]	[–0.043, 0.283]	[–0.137, 0.207]
CRV1 p -value	.168	.717	.143	.683
Wild-cluster-bootstrap- t p -value	.192	.750	.195	.689
Expected Gaokao years	1999–2010	1999–2010	1999–2010	1999–2010
Observations	5,888	5,888	5,888	5,888
Adopting-province N	3,894	3,894	3,894	3,894
Province clusters	28	28	28	28

Notes: For each displayed shift $s \in \{-3, -2, +2, +3\}$, province p 's assigned adoption year is $G_p^* = G_p + s$, pseudo event time is $k^* = k - s$, and the same Sun–Abraham event-study specification is re-estimated. Figure D.1 supplies the recorded-date ($s = 0$) benchmark, which is intentionally not repeated as a model column. Across columns, the respondent sample (expected Gaokao years 1999–2010), fixed-effects structure, and reform-wave aggregation weights from the recorded-date specification (2004: .599; 2005: .228; 2006: .173) are held fixed; only the assigned dates and resulting pseudo-event indicators change. Pseudo event time $k^* = -2$ is omitted, and the adoption-boundary mean averages $k^* = -1, 0, +1$. All three reform waves support the pseudo-reference and the three pseudo-boundary cells under every assigned-date shift. The displayed outer cells under $s = -3$ and $s = +3$ have partial wave support, and their fixed base weights are renormalized over the supporting waves. CRV1 inference clusters by inferred Gaokao-hukou province (28 clusters; $df = 27$); wild-cluster-bootstrap- t p -values use 9,999 null-imposed Rademacher replications clustered at the same level. Significance symbols refer to two-sided CRV1 tests; wild-cluster-bootstrap- t p -values are reported separately. [†] $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Table D.2. Mapping Hypothetical-Date Cells into Recorded Event Time

Assigned date shift, s	Pseudo-boundary cells (recorded k)	Pseudo reference (recorded k)	Interpretation
–3	–4, –3, –2	–5	Non-overlapping pre-implementation timing placebo
–2	–3, –2, –1	–4	Pre-implementation near-boundary diagnostic; overlaps the recorded focal window at $k=-1$
0	–1, 0, +1	–2	Recorded-date specification
+2	+1, +2, +3	0	Post-adoption localization diagnostic
+3	+2, +3, +4	+1	Post-adoption localization diagnostic

Notes: G_p denotes province p 's recorded adoption year, $G_p^* = G_p + s$ its assigned adoption year, and t denotes expected Gaokao year. Recorded and pseudo event time are therefore $k = t - G_p$ and $k^* = t - G_p^* = k - s$, respectively. The table maps the pseudo-boundary cells $k^* = -1, 0, +1$ and the omitted pseudo reference $k^* = -2$ into recorded event time. Negative values of s assign earlier adoption dates, and positive values assign later dates. The recorded institutional boundary remains between $k = -1$, the final national-paper cohort, and $k = 0$, the first province-set-paper cohort.

Appendix E: Sensitivity to Expected Gaokao Year and Inferred Gaokao-Hukou Province

Table E.1. Event-Study Estimates under Alternative Expected-Year and Gaokao-Hukou Assignments

DV	Perceived Social Fairness (1–5; higher values indicate greater perceived fairness)				
	Observed birth month	Missing birth month: birth year +19	Exclude CGSS 2017 and 2018	High-confidence assignment (Rules S1–S3)	Exclude conflicting province assignments
$k = -3$	−0.014 (0.101)	0.005 (0.096)	0.012 (0.103)	−0.029 (0.115)	0.066 (0.086)
$k = -2$	(reference)	(reference)	(reference)	(reference)	(reference)
$k = -1$	−0.215 [†] (0.106)	−0.186 [†] (0.091)	−0.174 (0.106)	−0.294** (0.093)	−0.162* (0.076)
$k = 0$	−0.233 (0.141)	−0.203 [†] (0.106)	−0.201 (0.136)	−0.258 [†] (0.148)	−0.236* (0.110)
$k = +1$	−0.315* (0.153)	−0.341** (0.121)	−0.303* (0.144)	−0.197 (0.123)	−0.181 [†] (0.093)
$k = +2$	−0.188 (0.137)	−0.123 (0.122)	−0.166 (0.137)	−0.209 (0.175)	−0.121 (0.117)
Adoption-boundary mean: $k = -1, 0, +1$	−0.254* (0.115)	−0.244* (0.093)	−0.226 [†] (0.110)	−0.250* (0.104)	−0.193* (0.073)
Earlier-cohort diagnostic: $H_0 : \beta_{-3} = 0$	.891	.962	.907	.803	.446
Expected Gaokao-year window	2001–2007	2001–2007	2001–2007	2001–2007	2001–2007
Observations	2,368	3,353	2,422	2,678	3,276
Adopting-province N	1,585	2,234	1,629	1,795	2,179
Province clusters	28	28	28	28	28
Gaokao-hukou province FE	Yes	Yes	Yes	Yes	Yes
Expected Gaokao year FE	Yes	Yes	Yes	Yes	Yes
Survey province $\times$ CGSS wave FE	Yes	Yes	Yes	Yes	Yes

Notes: Columns 1–3 probe uncertainty arising from expected-Gaokao-year assignment. Column 1 retains respondents with an observed birth month. Column 2 assigns birth year plus 19 rather than plus 18 when birth month is missing, rematches affected respondents to province-year policy cells, and reapplies the 2001–2007 window. Column 3 excludes CGSS 2017 and 2018, whose public files omit birth month. Columns 4–5 probe uncertainty in inferred Gaokao-hukou province assignment. Column 4 restricts assignments to Rules S1–S3, and column 5 excludes the 111 preferred-sample respondents whose usable sources identify different provinces. Each specification holds the preferred-sample reform-wave aggregation weights fixed rather than recomputing them for the restricted sample. Event $k = -2$ is omitted, and standard errors are clustered by inferred Gaokao-hukou province. [†] $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Appendix F: Covariate-Adjusted Event-Study Models

Table F.1. Event-Study Estimates with Demographic and Income Controls

DV	Perceived Social Fairness (1–5; higher values indicate greater perceived fairness)			
	Demographic controls		Demographic and income controls	
Model				
Event time	Estimate (SE)	Adopting-province N	Estimate (SE)	Adopting-province N
–3	0.042 (0.083)	320	0.020 (0.083)	287
–2	(reference)	293	(reference)	265
–1	–0.182* (0.073)	310	–0.210** (0.067)	278
0	–0.271* (0.108)	301	–0.296* (0.110)	279
+1	–0.200 [†] (0.098)	350	–0.174 (0.105)	318
+2	–0.153 (0.124)	328	–0.143 (0.118)	288
<i>Individual covariate estimates</i>				
Male	0.014 (0.023)		0.010 (0.028)	
Han ethnicity	–0.140 (0.084)		–0.127 (0.089)	
Chinese Communist Party member	0.038 (0.058)		0.053 (0.067)	
Annual individual income (10,000 yuan)	–		0.003 (0.003)	
Adoption-boundary mean: $k = -1, 0, +1$	–0.218** (0.074)	961	–0.227** (0.074)	875
Earlier-cohort diagnostic: $H_0 : \beta_{-3} = 0$	$p = .619$	320	$p = .812$	287
Individual controls	Yes		Yes	
Expected Gaokao-year window	2001–2007		2001–2007	
Observations	3,381		3,059	
Adopting-province N	2,256		2,036	
Province clusters	28		28	
Gaokao-province FE	Yes		Yes	
Expected-year FE	Yes		Yes	
Survey province $\times$ wave FE	Yes		Yes	

Notes: The first model includes indicators for male, Han ethnicity, and Chinese Communist Party membership and uses respondents observed on all three variables. The second model additionally includes annual individual income and therefore requires income to be observed; the two columns consequently use different complete-case samples. The individual-control coefficients and their clustered standard errors are reported below the event-study estimates. Income is measured in units of 10,000 yuan and is not logged. Covariates are measured at the time of the survey and included as a sensitivity check. Both models retain the event-study structure, fixed effects, and fixed preferred-sample reform-wave weights used in Table C.1. Event $k = -2$ is omitted. Adopting-province N reports unweighted respondent counts in the 14 provinces that adopted during 2004–2006. The adoption-boundary estimate is the equally weighted mean of the coefficients at $k = -1, 0, +1$, while its reported N sums the corresponding event-cell counts; the earlier-cohort row uses $k = -3$. FE denotes fixed effects. Standard errors, in parentheses, are clustered by inferred Gaokao-hukou province. [†] $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Appendix G: Outcome-Specificity Diagnostics

Table G.1. Event-Study Estimates for Companion Outcomes

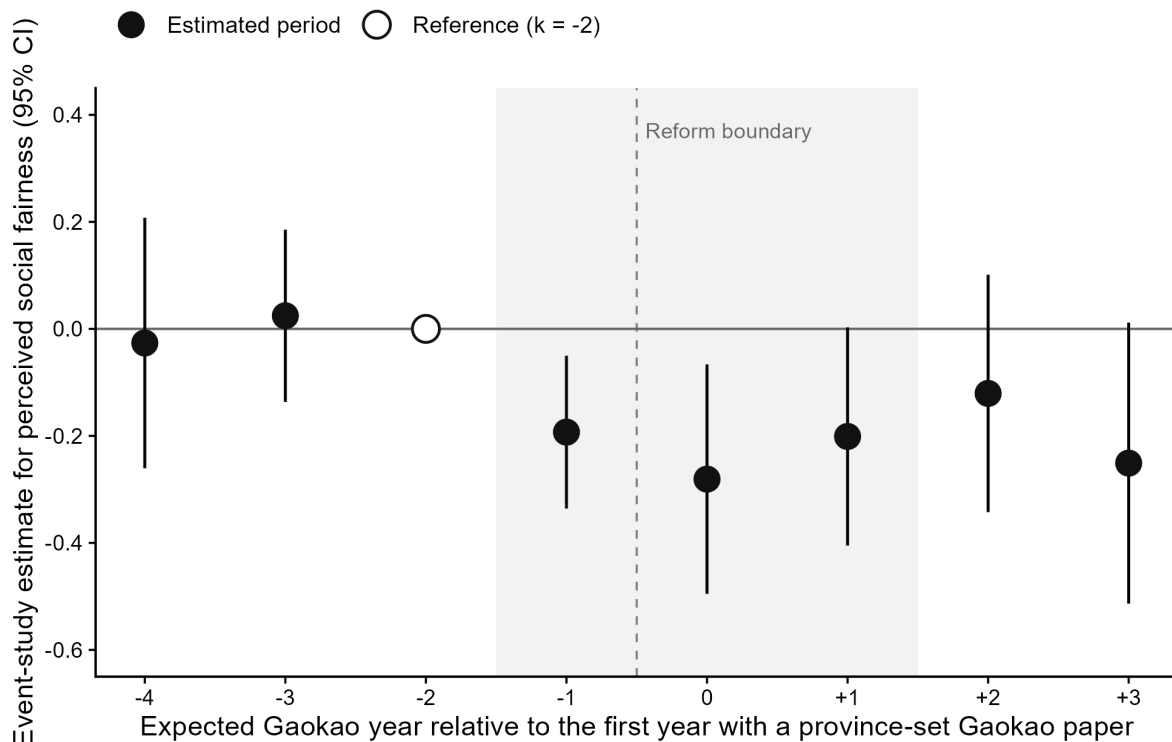
<i>Panel A. Companion-outcome event studies on maximal available samples</i>				
DV	Generalized Trust (1–5; higher = stronger agreement)		Expected Exploitation by Others (1–5; higher values indicate greater expected exploitation)	
Event time	Estimate (SE)	Adopting- province N	Estimate (SE)	Adopting- province N
–3	0.113 [†] (0.064)	322	–0.212* (0.102)	322
–2	(reference)	295	(reference)	293
–1	–0.093 (0.083)	311	0.027 (0.106)	310
0	–0.008 (0.093)	301	–0.002 (0.094)	301
+1	–0.099 (0.078)	349	0.104 (0.095)	348
+2	0.228* (0.098)	330	–0.041 (0.116)	329
Adoption-boundary mean: $k = -1, 0, +1$	–0.066 (0.066)	961	0.043 (0.073)	959
Earlier-cohort diagnostic: $H_0 : \beta_{-3} = 0$	$p = .089$	322	$p = .048$	322
Observations	3,391		3,384	
Adopting-province N	2,262		2,257	
Province clusters	28		28	
<i>Panel B. Pairwise Common-Sample Outcome Comparisons</i>				
Outcome	Generalized Trust		Reverse-Coded Expected Nonexploitation	
Observations	3,386		3,380	
Adopting-province N	2,259		2,256	
Province clusters	28		28	
Fairness boundary mean (raw scale)	–0.214** (0.073)		–0.221** (0.073)	
Companion-outcome boundary mean (raw scale)	–0.068 (0.065)		–0.046 (0.072)	
Fairness minus companion outcome (raw scale; 95% CI; p)	–0.146 [†] (0.084); [–0.318, 0.025]; $p = .092$		–0.175 [†] (0.097); [–0.373, 0.024]; $p = .082$	
Fairness boundary mean (SD units)	–0.211** (0.072)		–0.217** (0.071)	
Companion-outcome boundary mean (SD units)	–0.066 (0.063)		–0.046 (0.072)	
Fairness minus companion outcome (SD units; 95% CI; p)	–0.144 [†] (0.082); [–0.313, 0.025]; $p = .091$		–0.172 [†] (0.095); [–0.368, 0.024]; $p = .083$	

Notes: Generalized trust records agreement that most people can be trusted; expected exploitation records agreement that, if one is not careful, others will try to take advantage. Panel A retains the original coding of expected exploitation. For the cross-outcome comparison in Panel B, expected exploitation is reverse-coded as six minus the original response so that higher values on fairness, trust, and nonexploitation consistently indicate a more favorable social evaluation. Panel A uses the maximal available sample for each companion outcome; Panel B uses respondents jointly observed on fairness and that companion outcome. Raw and standardized differences are linear restrictions from stacked models using their full province-clustered covariance matrices and outcome-specific fixed effects. All models hold the preferred-sample reform-wave weights fixed, omit $k = -2$, and use CRV1 inference clustered by inferred Gaokao-hukou province. As in Table C.1, the Panel A estimates at $k = +2$ are supported only by the 2004 and 2005 adoption waves, with the fixed weights renormalized over those waves. [†] $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Appendix H: Window and Cohort-Trend Sensitivity

Figure H.1 plots the single-reference event study for expected Gaokao years 2000–2008. Panel A of Table H.1 examines sensitivity to the extended cohort window and to adjustment for a baseline college-enrollment-share trend. Panel B compares extended-window models with and without province-specific linear cohort trends under the same two-period normalization.

Figure H.1. Perceived-Fairness Event-Study Estimates, Expected Gaokao Years 2000–2008



Notes: Points are event-time coefficients from column 2 of Table H.1; bars show 95% confidence intervals based on CRV1 standard errors clustered by inferred Gaokao-hukou province. The open point at $k = -2$ is the omitted reference, the horizontal line marks zero, the vertical dashed line marks the reform boundary between $k = -1$ and $k = 0$, and the gray band marks the focal $k = -1, 0, +1$ adoption-boundary window. The $k = +3$ coefficient is supported only by the 2004 and 2005 adoption waves, so the fixed reform-wave weights are renormalized over those waves.

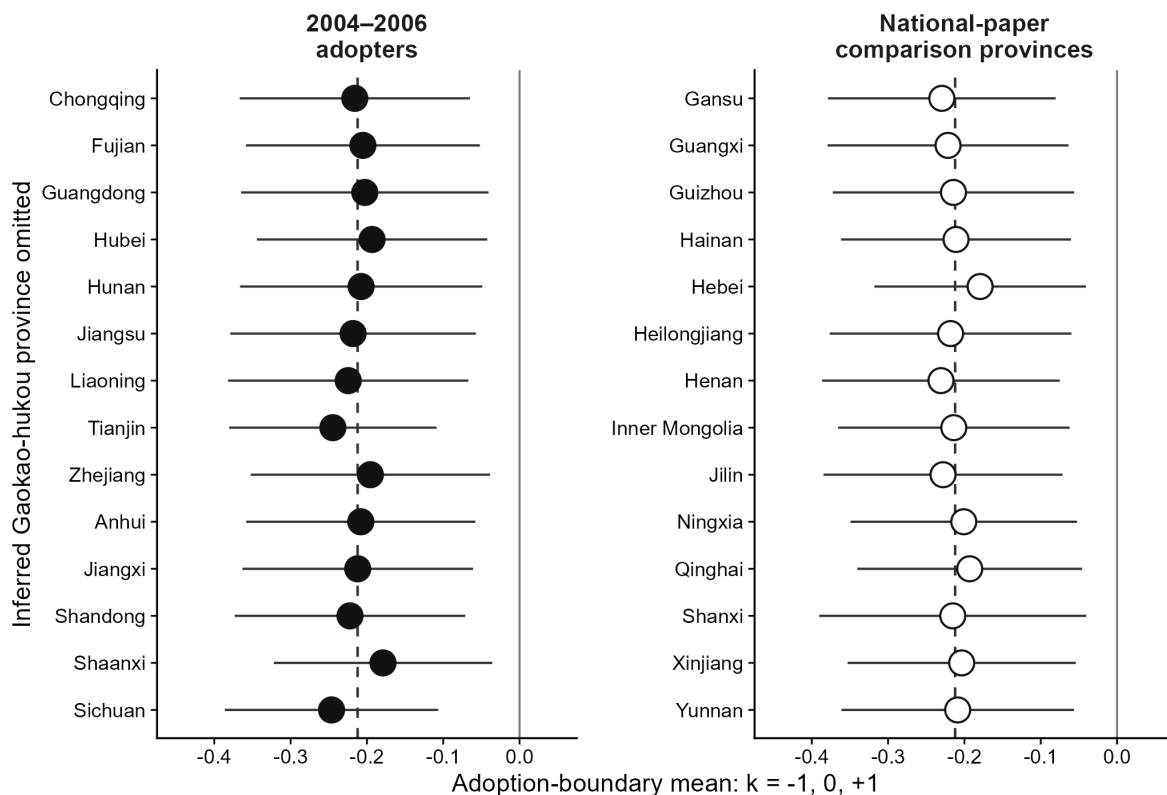
Table H.1. Event-Study Estimates for Window and Cohort-Trend Sensitivity

DV	Perceived Social Fairness (1–5; higher values indicate greater perceived fairness)			
	<i>Panel A. Window and baseline-opportunity trend sensitivity ($k = -2$)</i>		<i>Panel B. Matched province-trend diagnostic (anchored at $k = -3$ and $k = -2$)</i>	
Panel				
Event time	Preferred window + college-share trend	Single-reference extended-window event study	Extended window no province trends	Extended window province trends
$k = -4$	—	−0.026 (0.114)	−0.032 (0.102)	−0.067 (0.150)
$k = -3$	0.036 (0.087)	0.024 (0.078)	(reference)	(reference)
$k = -2$	(reference)	(reference)	(reference)	(reference)
$k = -1$	−0.172* (0.073)	−0.193** (0.070)	−0.180* (0.066)	−0.176 (0.118)
$k = 0$	−0.259* (0.098)	−0.281* (0.104)	−0.267* (0.105)	−0.241 (0.189)
$k = +1$	−0.190† (0.097)	−0.201† (0.099)	−0.200† (0.104)	−0.146 (0.264)
$k = +2$	−0.133 (0.121)	−0.121 (0.108)	−0.120 (0.111)	−0.044 (0.346)
$k = +3$	—	−0.251† (0.128)	−0.231† (0.113)	−0.269 (0.541)
<i>Linear trend adjustment</i>				
College share in 2000 × linear expected-Gaokao-year trend	−0.078 (0.324)	—	—	—
Normalization period(s)	$k = -2$	$k = -2$	$k = -3, -2$	$k = -3, -2$
Adoption-boundary mean: $k = -1, 0, +1$	−0.207** (0.068)	−0.225** (0.073)	−0.216** (0.076)	−0.188 (0.181)
Immediate-boundary mean: $k = -1, 0$	−0.216** (0.072)	−0.237** (0.075)	−0.224** (0.077)	−0.208 (0.147)
Post-adoption mean: $k = 0, +1$	−0.225* (0.084)	−0.241* (0.091)	−0.234* (0.095)	−0.194 (0.221)
Earlier cohort(s) tested	$k = -3$	$k = -4, -3$	$k = -4$	$k = -4$
Earlier-cohort diagnostic: joint p -value	.677	.877	.761	.657
Expected Gaokao-year window	2001–2007	2000–2008	2000–2008	2000–2008
Observations	3,387	4,491	4,491	4,491
Adopting-province N	2,260	2,988	2,988	2,988
Province clusters	28	28	28	28
Gaokao-hukou province FE	Yes	Yes	Yes	Yes
Expected Gaokao year FE	Yes	Yes	Yes	Yes
Survey province × CGSS wave FE	Yes	Yes	Yes	Yes
Province-specific linear cohort trends	No	No	No	Yes

Notes: Column 1 interacts the province's 2000 college-enrollment share with a linear trend in expected Gaokao year; both variables are mean-centered in the estimation sample. The share is the proportion, ranging from 0 to 1, of 19–21-year-olds enrolled in regular junior-college or bachelor's programs, constructed from the 2000 Chinese Population Census 1 percent long-form sample. Columns 1 and 2 omit $k = -2$. Columns 3 and 4 omit $k = -3$ and $k = -2$ because the province-trend model requires level and slope anchors; their only difference is the province-specific trends, so column 4 should be compared with column 3. The three mean rows equally average $k = -1, 0, +1$, $k = -1, 0$, and $k = 0, +1$, respectively, using the complete province-clustered covariance matrix. All columns use the fixed reform-wave weights; partial-support outer events renormalize them over supported waves. Standard errors are clustered by inferred Gaokao-hukou province. † $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Appendix I: Leave-One-Province-Out Sensitivity

Figure I.1. Leave-One-Province-Out Estimates of the Adoption-Boundary Mean



Notes: Each point re-estimates the preferred specification after omitting the named inferred Gaokao-hukou province. The panels classify the omitted province by its role in the preferred sample. Bars show 95% confidence intervals based on CRV1 standard errors clustered by the 27 remaining provinces. The preferred-sample reform-wave aggregation weights are held fixed across omissions; only observations from the named province are removed. The dashed line marks the preferred-sample estimate of -0.212 using all 28 provinces. Leave-one-province-out estimates range from -0.246 to -0.179.

Appendix J: Heterogeneity by Pre-Reform Undergraduate-Institution Density

Table J.1. Event-Study Comparison by Pre-Reform Undergraduate-Institution Density

DV	Perceived Social Fairness (1–5; higher values indicate greater perceived fairness)		
	Lower-density 2004 adopters	Higher-density 2004 adopters	Lower-minus-higher difference
<i>Panel A. Group-Specific Event Paths and Descriptive Differences</i>			
$k = -3$	–0.097 (0.182)	0.083 (0.125)	–0.180 (0.140)
$k = -2$	(reference)	(reference)	(reference)
$k = -1$	–0.358 [†] (0.195)	–0.185 (0.149)	–0.172 (0.187)
$k = 0$	–0.720*** (0.109)	–0.057 (0.150)	–0.663*** (0.134)
$k = +1$	–0.384* (0.177)	–0.059 (0.124)	–0.325 [†] (0.180)
$k = +2$	–0.276* (0.127)	0.085 (0.135)	–0.362** (0.098)
Adoption-boundary mean: $k = -1, 0, +1$	–0.487*** (0.096)	–0.100 (0.113)	–0.387*** (0.094)
<i>Panel B. Focused Contrasts and Inference</i>			
Adoption-boundary mean: $k = -1, 0, +1$	–0.487 (0.096)	–0.100 (0.113)	–0.387 (0.094) 95% CRV1 CI [–0.580, –0.193] CRV1 $p < .001$ wild-bootstrap- t $p = .005$
Adoption-year contrast: $k = 0$	–0.720 (0.109)	–0.057 (0.150)	–0.663 (0.134) 95% CRV1 CI [–0.937, –0.389] CRV1 $p < .001$ wild-bootstrap- t $p = .005$
Earlier-cohort diagnostic: $k = -3$	–0.097 (0.182)	0.083 (0.125)	–0.180 (0.140) 95% CRV1 CI [–0.468, 0.108] CRV1 $p = .211$ wild-bootstrap- t $p = .309$
Expected Gaokao-year window		2001–2007	
Observations		3,387	
Focal adopter provinces	4	4	—
Focal-group observations in full joint model	622	571	—
All-adopter observations		2,260	
Province clusters		28	

Notes: Table J.2 defines the groups. All estimates come from one joint model of the full preferred sample, with separate event paths for the focal groups and nuisance adopters, province, expected-*gaokao*-year, and survey-context fixed effects, and $k = -2$ omitted. Median-ranked Guangdong and all 2005/2006 adopters enter the nuisance path. Panel A reports group-specific Sun–Abraham event-time coefficients and lower-minus-higher descriptive differences; parentheses contain CRV1 standard errors clustered by inferred *gaokao*-hukou province. Panel B reports the three focused contrasts and adds 95% CRV1 confidence intervals, exact CRV1 p -values, and wild-cluster-bootstrap- t p -values based on 9,999 null-imposed Rademacher replications. Both focal groups adopted in 2004, so separate common-wave reweighting is redundant. Reported confidence intervals and p -values are unadjusted for the post hoc choice of the heterogeneity measure and grouping. Replacing Zhejiang with Guangdong in the lower-density group yields an adoption-boundary lower-minus-higher difference of –0.353 (95% CRV1 CI [–0.562, –0.144]; CRV1 $p = .002$; wild-cluster-bootstrap- t $p = .004$). Significance symbols correspond to two-sided CRV1 tests: [†] $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Table J.2. Construction of Undergraduate-Institution-Density Groups among 2004 Adopters

<i>Panel A. 2004 Reform-Cohort Ranking and Group Assignment</i>					
Province	2003 Gaokao candidates	Ordinary undergraduate institutions	Institutions per 100,000 candidates	Density rank	Density group
Hunan	~300,000	22	7.33	1	Lower-density
Fujian	190,837	14	7.34	2	Lower-density
Hubei	~330,000	31	9.39	3	Lower-density
Zhejiang	~229,000	23	10.04	4	Lower-density
Guangdong	~306,400	31	10.12	5	
Jiangsu	~341,000	42	12.32	6	Higher-density
Chongqing	94,539	16	16.92	7	Higher-density
Liaoning	~176,000	38	21.59	8	Higher-density
Tianjin	57,797	18	31.14	9	Higher-density

<i>Panel B. Measure Construction</i>	
Component	Definition
Candidate denominator	Reported 2003 <i>gaokao</i> candidates; ~ marks an approximate source value.
Institution numerator	Ordinary undergraduate institutions located in the province in 2003.
Density and groups	Institutions divided by candidates, multiplied by 100,000; ranks 1–4 are lower-density and ranks 6–9 are higher-density.

<i>Panel C. Candidate-Count Sources</i>	
Province	Source report(s)
Hunan	Sina Education (2003-11-17)
Fujian	China Education Online (2003-05-07)
Hubei	China Youth Daily (2007-04-08)
Zhejiang	Sina Education (2003-05-19)
Guangdong	Sina Education (2003-06-16)
Jiangsu	Sina Education (2009-06-01)
Chongqing	China Education Online (2003-05-03); China Education Online (2003-05-13)
Liaoning	Sina Education (2003-06-08)
Tianjin	Sina Education (2003-11-17); China Education Online (2003-05-15); CCTV (2004-04-06)

Notes: The ranking covers the nine provinces that adopted province-set papers in 2004. Density and group assignment are defined in Panel B; Guangdong, ranked fifth, is left unassigned. Panel C links the reports used for the candidate counts. Institution counts for all provinces come from the 2003 China Education Statistical Yearbook.

Table J.3. Focal Leave-One-Province-Out Diagnostics

Omitted province	Original density group	Lower-minus-higher difference	CRV1 SE	95% CRV1 CI	Wild p
Hunan	Lower-density	−0.436	0.090	[−0.621, −0.252]	.013
Fujian	Lower-density	−0.384	0.098	[−0.585, −0.183]	.009
Hubei	Lower-density	−0.351	0.093	[−0.542, −0.160]	.029
Zhejiang	Lower-density	−0.398	0.109	[−0.623, −0.173]	.024
Jiangsu	Higher-density	−0.471	0.080	[−0.635, −0.308]	.010
Chongqing	Higher-density	−0.377	0.096	[−0.575, −0.179]	.009
Liaoning	Higher-density	−0.399	0.127	[−0.661, −0.137]	.008
Tianjin	Higher-density	−0.316	0.075	[−0.470, −0.162]	.012

Notes: Each row deletes all records for the named province and re-estimates the full joint specification for the adoption-boundary lower-minus-higher difference. Guangdong remains on the nuisance path and does not replace the omitted province. CRV1 standard errors and confidence intervals use the 27 remaining province clusters; the confidence intervals are unadjusted. Wild-cluster-bootstrap- t p -values are based on 9,999 null-imposed Rademacher replications clustered at the same level and are also unadjusted.

Appendix K: Survey Design, Recruitment, and Survey Flow

Study 2 was a two-arm, between-subjects online survey experiment fielded August 18–20, 2026. An online provider recruited adults residing in mainland China. The preregistered plan imposed no demographic or soft quotas. The provider used unique user IDs intended to permit one response per respondent; it supplied no count of duplicate or blocked attempts.

The preregistered pooled target was 2,400 completed, AC0-passing randomized questionnaires, with minor overshoot retained. Assuming equal allocation and 2,400 nonmissing O1 observations, the preregistered normal-approximation calculation yielded minimum detectable effects of approximately 0.114 and 0.132 standard deviations at 80% and 90% power, respectively, for a two-sided difference-in-means test at $\alpha = .05$. For field administration, the provider collected 2,450 completed questionnaires before applying the AC0 delivery screen. It removed 37 completed questionnaires with incorrect AC0 responses before data delivery. The delivered file therefore contained 2,413 completed questionnaires: 1,206 assigned to the nationally uniform-paper condition and 1,207 assigned to the province-specific-paper condition. The researchers retained all 2,413 records, including the preregistered 13-case overshoot.

In the questionnaire, informed consent and AC0 preceded equal-probability assignment. Respondents then read the vignette; answered O1–O8 and O9a–O9d; completed an analytically separate list experiment, D1–D3, AC1, and D4–D14; and received a debrief. Navigation was forward-only. Respondents could answer a question and continue, skip it, or exit the survey at any point. Once they advanced, they could not return to earlier questions to review or revise their responses. Figure 5 in the main text summarizes participant disposition (Panel A) and this survey sequence (Panel B).

No respondent-level records for the 37 AC0 removals were delivered. The provider supplied no separate randomization denominator or records or counts for incomplete attempts. The 2,450 count is an administrative total of completed questionnaires, not a survey-start or

randomization denominator; overall and arm-specific completion or attrition rates cannot be assessed. Analyses compare assignments within the delivered sample. No post-treatment attention, comprehension, list-module response, or completion-time measure was an exclusion criterion in the preregistered analyses. Outcome models use outcome-specific nonmissing samples; the D6 and D8 models additionally require an observed value on the relevant background item.

Appendix L: Field Instrument, Measures, Coding, and Ethical Considerations

Both written vignettes described the same national talent-selection program, with the same administering body, a common examination syllabus and maximum score, province-varying quotas, selection chances, and cutoffs, and descending score-based selection within each province. They differed only in whether the administering body prepared one examination paper for use nationwide (A1) or separate papers for each province (A2). Table L.1 presents the original Chinese wording and English translations of the outcome and comprehension items, together with their response scales, coding, and preregistered analysis roles.

Participation was voluntary and based on informed consent. Each participant received US\$0.50 for completing the survey, which was estimated to take 6–8 minutes. The questionnaire did not request directly identifying information, and data were analyzed in de-identified form with results reported in aggregate. The debrief explained that the talent-selection program was a hypothetical scenario created for the study and was not a real or ongoing government program. The author’s institutional review board determined the study to be exempt on May 14, 2026. The pre-analysis plan (PAP) was registered on OSF before survey fieldwork began on August 18, 2026.

Full Treatment Vignettes as Fielded

The Chinese vignettes below reproduce the wording and emphasis of the final questionnaire. English translations are provided for reference. Respondents saw only the Chinese text of their assigned condition.

Condition A1: Nationally Uniform Paper

Original Chinese wording

设想国家正在实施一项全国青年人才选拔计划，面向全国招募一批青年储备人才，进入中央部委和中央企业在各地设立的初级岗位工作。

申请者需要提交材料，并参加考试选拔。考试的组织、监考和阅卷均由该计划的管理机构负责。

这项计划采取**分省录取**：有的省份分配到的名额较多，入选机会相对更大；有的省份分配到的名额较少，入选机会相对更小。在每个省份内部，按照申请者的考试成绩从高到低录取。**不同省份的录取分数线并不相同。**

管理机构依据全国统一的考试大纲，**为全国申请者统一命制一套试卷。**

各省考试满分相同，并使用相同的试卷。

English translation

Imagine that the state is implementing a National Young Talent Selection Program to recruit a group of young reserve personnel nationwide to work in entry-level positions established around the country by central government ministries and centrally administered state-owned enterprises.

Applicants are required to submit application materials and take an examination. The program's administrative body is responsible for organizing, proctoring, and grading the examination.

The program **selects candidates separately by province**: some provinces are allocated more positions, so applicants there have a relatively greater chance of being selected; other provinces are allocated fewer positions, so applicants there have a relatively smaller chance of being selected. Within each province, applicants are selected from the highest to the lowest examination scores. **Cutoff scores differ across provinces.**

Based on a nationally uniform examination syllabus, the administrative body **develops one uniform exam paper for applicants nationwide.**

The maximum score is the same in every province, and all provinces use the same exam paper.

Condition A2: Province-Specific Papers

Original Chinese wording

设想国家正在实施一项全国青年人才选拔计划，面向全国招募一批青年储备人才，进入中央部委和中央企业在各地设立的初级岗位工作。

申请者需要提交材料，并参加考试选拔。考试的组织、监考和阅卷均由该计划的管理机构负责。

这项计划采取**分省录取**：有的省份分配到的名额较多，入选机会相对更大；有的省份分配到的名额较少，入选机会相对更小。在每个省份内部，按照申请者的考试成绩从高到低录取。**不同省份的录取分数线并不相同。**

管理机构依据全国统一的考试大纲，**分别为各省命制试卷。**

各省考试满分相同，但不同省份使用不同的试卷。

English translation

Imagine that the state is implementing a National Young Talent Selection Program to recruit a group of young reserve personnel nationwide to work in entry-level positions established around the country by central government ministries and centrally administered state-owned enterprises.

Applicants are required to submit application materials and take an examination. The program's administrative body is responsible for organizing, proctoring, and grading the examination.

The program **selects candidates separately by province**: some provinces are allocated more positions, so applicants there have a relatively greater chance of being selected; other provinces are allocated fewer positions, so applicants there have a relatively smaller chance of being selected. Within each province, applicants are selected from the highest to the lowest examination scores. **Cutoff scores differ across provinces.**

Based on a nationally uniform examination syllabus, the administrative body **develops a separate exam paper for each province.**

The maximum score is the same in every province, but different provinces use different exam papers.

Table L.1. Bilingual Survey Items, Coding, and Analysis Roles

Item and analysis role	Chinese wording	English translation	Response scale and coding
<i>Panel A. Outcome items in questionnaire order</i>			
O1 Sole confirmatory outcome	假设一个社会主要通过这样的制度来选拔人才，并分配重要的发展机会。总的来说，您认为这样的社会公平吗？	Suppose a society primarily uses a system like this to select talent and allocate important opportunities for advancement. Overall, do you think such a society is fair?	1 = Very unfair; 5 = Very fair.
O2 Preregistered supporting outcome	总体来说，您认为这种选拔制度公平吗？	Overall, do you think this selection system is fair?	1 = Very unfair; 5 = Very fair.
O3a Construct validation	在这种制度下，不同省份申请者的考试成绩可以相互比较。	Under this system, the examination scores of applicants from different provinces can be compared with one another.	1 = Strongly disagree; 5 = Strongly agree. Optional; no DK/PNA option.
O3b Construct validation	即使申请者来自不同省份，也可以根据考试成绩判断谁表现得更好。	Even when applicants come from different provinces, their examination scores can be used to determine who performed better.	1 = Strongly disagree; 5 = Strongly agree. Optional; no DK/PNA option.
O4 Discriminant validity	在每个省份内部，这项考试的成绩在多大程度上能够准确反映本省申请者的能力和努力？	Within each province, to what extent can the examination scores accurately reflect the ability and effort of applicants from that province?	1 = Not at all; 5 = Completely. Conventional contrast and prespecified TOST.
O5 Preregistered exploratory diagnostic	您认为这项计划在不同省份之间分配录取名额的方式公平吗？	Do you think the way this program allocates selection quotas across provinces is fair?	1 = Very unfair; 5 = Very fair.
O6 Preregistered exploratory diagnostic	在这种制度下，普通人是否可以通过努力获得上升机会？	Under this system, can ordinary people obtain opportunities for upward mobility through hard work?	1 = Definitely cannot; 3 = Hard to say; 5 = Definitely can.
O7 Preregistered exploratory diagnostic	您认为在这种选拔制度下，考试和录取过程中出现违规或舞弊行为的可能性有多大？	Under this selection system, how likely do you think it is that rule violations or cheating will occur during the examination and selection process?	1 = Very unlikely; 5 = Very likely; higher values indicate more risk.
O8 Preregistered exploratory diagnostic	您认为，这项计划采用的试卷设置方式，对来自不同省份的申请者来说公平吗？	Do you think the exam-paper arrangement used by this program is fair to candidates from different provinces?	1 = Very unfair; 5 = Very fair.

Continued on next page

Table L.1 continued			
Item and analysis role	Chinese wording	English translation	Response scale and coding
Panel B. Factual manipulation and comprehension items			
O9a Treatment-reception diagnostic	根据刚才的材料，不同省份使用的考试试卷是如何安排的？	According to the material you just read, how were the exam papers used by different provinces arranged?	The correct answer varies by arm.
O9b Information-equivalence diagnostic	根据刚才的材料，各省考试的大纲和总分是如何安排的？	According to the material you just read, how were the examination syllabus and maximum score arranged across provinces?	The correct answer is common to both arms.
O9c Information-equivalence diagnostic	根据刚才的材料，这项计划的录取名额如何安排？	According to the material you just read, how were the selection quotas for this program arranged?	The correct answer is common to both arms.
O9d Information-equivalence diagnostic	根据刚才的材料，在每个省份内部，申请者按照什么规则录取？	According to the material you just read, what rule was used to select applicants within each province?	The correct answer is common to both arms.

Notes: For O1, O2, and O4–O8, DK/PNA and nonresponse are coded missing, and each analysis uses the outcome-specific nonmissing sample. O3 is the mean of O3a and O3b when both items are observed; the two optional items have no separate DK/PNA option, and item-level analyses are also retained. For O9a–O9d, a correct answer is coded 1, any selected incorrect answer including “not mentioned” or “unsure” is coded 0, and nonresponse is missing.

Table L.2. Screening, Attention, and Background Items

Item(s)	Purpose	Coding and analytic use
AC0	Pretreatment reading-confirmation and eligibility screen	The post office was correct. The item preceded random assignment. The provider removed completed questionnaires with an incorrect response before data delivery; those records are absent from the delivered analysis file.
AC1	Post-treatment reading-attention and data-quality diagnostic	The post office was coded correct. Reported only; never used to exclude a record, redefine treatment, or define compliance.
D1–D14	Demographic and background measures	Prefer-not-to-answer and blank responses are coded missing. “Do not know / not sure” responses to parental-education items and “do not know / cannot remember clearly” responses to the hukou item are retained for descriptive summaries but treated as missing when the underlying variable enters a model. Elicited after assignment and used descriptively or in exploratory analyses; no causal moderation is identified.

Notes: D1–D14 cover age, gender, education, parental education, gaokao participation and age-16 hukou, public-sector-examination participation, income, residence, employment, family economic circumstances before adulthood, relationship status, and political affiliation.

Appendix M: Sample Composition and Outcome Availability

Table M.1 describes the completed sample by treatment arm. Table M.2 reports outcome missingness and completed-sample extreme-value sensitivity bounds for O1 and O7, and Figure M.1 shows the age-16 hukou distribution among respondents who reported taking the gaokao.

Table M.1. Descriptive Composition of the Completed Study 2 Sample by Treatment Arm

Characteristic / category	Overall ($N = 2,413$ n (%))	A1 ($N = 1,206$ n (%))	A2 ($N = 1,207$ n (%))	SMD
Age				
18–24	680 (28.2)	347 (28.8)	333 (27.6)	-0.026
25–34	964 (40.0)	485 (40.2)	479 (39.7)	-0.011
35–44	539 (22.3)	257 (21.3)	282 (23.4)	0.049
45–54	152 (6.3)	74 (6.1)	78 (6.5)	0.013
55 or older	75 (3.1)	42 (3.5)	33 (2.7)	-0.043
Missing / prefer not to answer	3 (0.1)	1 (0.1)	2 (0.2)	0.023
Gender				
Female	1,540 (63.8)	785 (65.1)	755 (62.6)	-0.053
Male	869 (36.0)	418 (34.7)	451 (37.4)	0.056
Missing / prefer not to answer	4 (0.2)	3 (0.2)	1 (0.1)	-0.041
Education				
Junior middle school or less	13 (0.5)	8 (0.7)	5 (0.4)	-0.034
High school / secondary technical school	69 (2.9)	35 (2.9)	34 (2.8)	-0.005
Vocational or junior college	225 (9.3)	107 (8.9)	118 (9.8)	0.031
Bachelor's degree	1,749 (72.5)	871 (72.2)	878 (72.7)	0.012
Master's degree	332 (13.8)	172 (14.3)	160 (13.3)	-0.029
Doctoral degree	23 (1.0)	12 (1.0)	11 (0.9)	-0.009
Missing / prefer not to answer	2 (0.1)	1 (0.1)	1 (0.1)	0.000
Father's education				
Junior middle school or less	872 (36.1)	446 (37.0)	426 (35.3)	-0.035
High school / secondary technical school	819 (33.9)	401 (33.3)	418 (34.6)	0.029
Vocational or junior college	376 (15.6)	187 (15.5)	189 (15.7)	0.004
Bachelor's degree	314 (13.0)	157 (13.0)	157 (13.0)	0.000
Master's degree	15 (0.6)	4 (0.3)	11 (0.9)	0.074
Doctoral degree	2 (0.1)	2 (0.2)	0 (0.0)	-0.058
Do not know / not sure	15 (0.6)	9 (0.7)	6 (0.5)	-0.032
Mother's education				
Junior middle school or less	1,110 (46.0)	558 (46.3)	552 (45.7)	-0.011
High school / secondary technical school	711 (29.5)	363 (30.1)	348 (28.8)	-0.028
Vocational or junior college	321 (13.3)	155 (12.9)	166 (13.8)	0.027
Bachelor's degree	242 (10.0)	117 (9.7)	125 (10.4)	0.022
Master's degree	14 (0.6)	6 (0.5)	8 (0.7)	0.022
Doctoral degree	2 (0.1)	1 (0.1)	1 (0.1)	0.000
Do not know / not sure	13 (0.5)	6 (0.5)	7 (0.6)	0.011
Gaokao experience				
Did not take the gaokao	58 (2.4)	30 (2.5)	28 (2.3)	-0.011
Took the gaokao	2,351 (97.4)	1,174 (97.3)	1,177 (97.5)	0.011
Missing / prefer not to answer	4 (0.2)	2 (0.2)	2 (0.2)	0.000
Public-sector exam experience				
Did not take a public-sector employment exam	1,151 (47.7)	583 (48.3)	568 (47.1)	-0.026
Took a public-sector employment exam	1,240 (51.4)	612 (50.7)	628 (52.0)	0.026
Missing / prefer not to answer	22 (0.9)	11 (0.9)	11 (0.9)	0.000
Monthly personal income				
No personal income	142 (5.9)	76 (6.3)	66 (5.5)	-0.035
1–2,999 yuan	258 (10.7)	130 (10.8)	128 (10.6)	-0.006
3,000–5,999 yuan	617 (25.6)	301 (25.0)	316 (26.2)	0.028
6,000–9,999 yuan	777 (32.2)	395 (32.8)	382 (31.6)	-0.024
10,000–14,999 yuan	392 (16.2)	189 (15.7)	203 (16.8)	0.031

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Table M.1 continued

Characteristic / category	Overall ($N = 2,413$ n (%))	A1 ($N = 1,206$ n (%))	A2 ($N = 1,207$ n (%))	SMD
15,000 yuan or more	207 (8.6)	109 (9.0)	98 (8.1)	-0.033
Missing / prefer not to answer	20 (0.8)	6 (0.5)	14 (1.2)	0.073
Current place of residence				
Rural village	127 (5.3)	62 (5.1)	65 (5.4)	0.011
Township	161 (6.7)	80 (6.6)	81 (6.7)	0.003
County seat or county-level city	578 (24.0)	293 (24.3)	285 (23.6)	-0.016
Small or medium-sized city	615 (25.5)	299 (24.8)	316 (26.2)	0.032
Large city	925 (38.3)	467 (38.7)	458 (37.9)	-0.016
Missing / prefer not to answer	7 (0.3)	5 (0.4)	2 (0.2)	-0.046
Employment status				
Employed full time	1,824 (75.6)	915 (75.9)	909 (75.3)	-0.013
Employed part time	77 (3.2)	39 (3.2)	38 (3.1)	-0.005
Student	425 (17.6)	208 (17.2)	217 (18.0)	0.019
Not currently employed	40 (1.7)	22 (1.8)	18 (1.5)	-0.026
Retired	13 (0.5)	5 (0.4)	8 (0.7)	0.034
Other	19 (0.8)	10 (0.8)	9 (0.7)	-0.009
Missing / prefer not to answer	15 (0.6)	7 (0.6)	8 (0.7)	0.010
Family economic circumstances before adulthood				
Relatively poor	239 (9.9)	122 (10.1)	117 (9.7)	-0.014
Lower middle	522 (21.6)	272 (22.6)	250 (20.7)	-0.045
Middle	1,216 (50.4)	601 (49.8)	615 (51.0)	0.022
Upper middle	299 (12.4)	152 (12.6)	147 (12.2)	-0.013
Relatively good	129 (5.3)	55 (4.6)	74 (6.1)	0.070
Missing / prefer not to answer	8 (0.3)	4 (0.3)	4 (0.3)	0.000
Relationship status				
Single	750 (31.1)	356 (29.5)	394 (32.6)	0.068
Stable relationship, not cohabiting	304 (12.6)	167 (13.8)	137 (11.4)	-0.075
Cohabiting	73 (3.0)	45 (3.7)	28 (2.3)	-0.082
Married	1,257 (52.1)	625 (51.8)	632 (52.4)	0.011
Divorced or separated	17 (0.7)	8 (0.7)	9 (0.7)	0.010
Widowed	1 (0.0)	1 (0.1)	0 (0.0)	-0.041
Missing / prefer not to answer	11 (0.5)	4 (0.3)	7 (0.6)	0.037
Political affiliation				
Chinese Communist Party member	483 (20.0)	243 (20.1)	240 (19.9)	-0.007
Communist Youth League member	663 (27.5)	329 (27.3)	334 (27.7)	0.009
Member of another political party	8 (0.3)	3 (0.2)	5 (0.4)	0.029
No party affiliation	1,240 (51.4)	621 (51.5)	619 (51.3)	-0.004
Other	7 (0.3)	2 (0.2)	5 (0.4)	0.046
Missing / prefer not to answer	12 (0.5)	8 (0.7)	4 (0.3)	-0.047

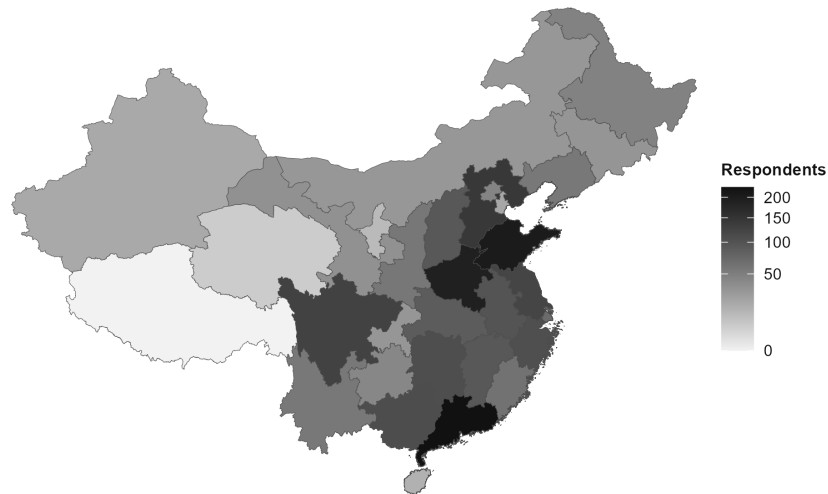
Notes: Entries are counts, with percentages in parentheses, for D1–D6 and D8–D14. Percentages use the full column sample as the denominator; percent signs are omitted. D7 (hukou province at age 16) is summarized among the 2,351 respondents who reported taking the gaokao in Figure M.1. All items were elicited after treatment assignment, and comparisons are descriptive of the delivered completed sample. SMD denotes the standardized mean difference for each category indicator, calculated as A2 (province-specific papers) minus A1 (nationally uniform paper). All reported absolute SMDs are below 0.10, used here only as a descriptive reference.

Table M.2. Outcome Missingness and Completed-Sample Sensitivity Bounds

<i>Panel A. Outcome missingness by treatment arm</i>			
Outcome	A1 missing <i>n</i> (%)	A2 missing <i>n</i> (%)	Difference (pp)
O1 Societal fairness	2 (0.2)	1 (0.1)	-0.08
O2 Selection-system fairness	1 (0.1)	2 (0.2)	0.08
O3 Comparability index	1 (0.1)	1 (0.1)	0.00
O3a Score comparability	0 (0.0)	0 (0.0)	0.00
O3b Rank comparability	1 (0.1)	1 (0.1)	0.00
O4 Within-province accuracy	2 (0.2)	2 (0.2)	0.00
O5 Quota-allocation fairness	7 (0.6)	8 (0.7)	0.08
O6 Upward mobility	6 (0.5)	4 (0.3)	-0.17
O7 Rule violations or cheating	41 (3.4)	25 (2.1)	-1.33
O8 Paper-arrangement fairness	3 (0.2)	6 (0.5)	0.25

<i>Panel B. Extreme-value sensitivity to item nonresponse</i>		
Outcome	Available-case A2–A1	Completed-sample extreme-value bounds
O1 Societal fairness	-0.1515	[-0.1561, -0.1461]
O7 Rule violations or cheating	0.0777	[-0.0418, 0.1770]

Notes: A1 contains 1,206 completed records and A2 contains 1,207. Panel A's difference is A2 minus A1 in percentage points, calculated from unrounded proportions; percent signs are omitted from body cells. Outcome analyses use outcome-specific nonmissing samples, without imputation or cross-outcome listwise deletion. Panel B replaces every missing response with the scale endpoints in the two arm-specific combinations least and most favorable to the A2–A1 mean difference. These deterministic bounds concern item nonresponse within the completed sample; they are not confidence intervals and do not address randomized noncompleters. All 66 missing O7 values are recorded don't-know/refusal responses, not blanks. O7 missingness was 3.40% in A1 and 2.07% in A2; the post hoc A2–A1 difference is -1.328 percentage points (HC2 95% CI [-2.630, -0.027]; nominal $p = .046$).

Figure M.1. Age-16 Hukou Provinces among Gaokao Takers

Notes: The map includes all 2,351 respondents who reported taking the gaokao. All were mapped to a mainland province-level unit; 30 of 31 units are represented, and Tibet has zero respondents. The map describes geographic coverage in the completed online sample, not population representativeness.

Appendix N: Preregistered Outcome Analyses

Table N.1 reports the preregistered outcome estimates and, in Panel D, the prespecified equivalence test for within-province evaluative accuracy (O4).

Table N.1. Preregistered Outcome Estimates and O4 Equivalence Test

	A1: Nationally uniform paper		A2: Province-specific papers					
Outcome	<i>N</i>	Mean (SD)	<i>N</i>	Mean (SD)	Difference (HC2 SE)	95% CI		<i>p</i>
<i>Panel A. Confirmatory and supporting outcomes</i>								
O1 Overall societal fairness	1,204	3.350 (1.066)	1,206	3.198 (1.070)	-0.151 (0.044)	[-0.237, -0.066]		< .001
O2 Fairness of the selection system	1,205	3.374 (1.081)	1,205	3.198 (1.100)	-0.176 (0.044)	[-0.263, -0.089]		< .001
<i>Panel B. Construct- and discriminant-validity measures</i>								
O3 Cross-province score comparability	1,205	3.564 (1.057)	1,206	2.110 (0.888)	-1.454 (0.040)	[-1.532, -1.376]		< .001
O3a Score comparability across provinces	1,206	3.460 (1.129)	1,207	2.047 (0.896)	-1.413 (0.041)	[-1.494, -1.332]		< .001
O3b Cross-province rank comparability	1,205	3.666 (1.155)	1,206	2.173 (1.061)	-1.493 (0.045)	[-1.582, -1.405]		< .001
O4 Within-province evaluative accuracy	1,204	4.008 (0.746)	1,205	3.901 (0.792)	-0.107 (0.031)	[-0.169, -0.046]		< .001
<i>Panel C. Preregistered exploratory diagnostic outcomes</i>								
O5 Quota-allocation fairness	1,199	3.051 (1.120)	1,199	2.960 (1.111)	-0.091 (0.046)	[-0.180, -0.002]		.046
O6 Opportunity for upward mobility	1,200	4.000 (0.846)	1,203	3.893 (0.887)	-0.107 (0.035)	[-0.177, -0.038]		.002
O7 Rule-violation or cheating risk	1,165	2.366 (1.071)	1,182	2.443 (1.103)	0.078 (0.045)	[-0.010, 0.166]		.084
O8 Paper-arrangement fairness	1,203	3.538 (1.102)	1,201	3.076 (1.139)	-0.462 (0.046)	[-0.552, -0.372]		< .001
<i>Panel D. Preregistered O4 equivalence test</i>								
Analysis	A1 SD	Equivalence bounds		90% CI		<i>p_L</i>	<i>p_U</i>	Conclusion
O4 equivalence test	0.746	[-0.112, 0.112]		[-0.159, -0.055]		.438	< .001	Not established

Notes: Panels A–C report A2 (province-specific papers) minus A1 (nationally uniform paper) differences from unweighted OLS without covariates, using the nonmissing sample for each outcome. Parentheses contain arm SDs and HC2 standard errors; 95% confidence intervals and two-sided *p*-values are unadjusted. O1 is the sole confirmatory outcome (descriptive Glass's $\delta = -0.142$); O2 is supporting; O3 and O4 are the prespecified nonconfirmatory construct- and discriminant-validity measures, respectively; O5–O8 are preregistered exploratory diagnostics; and O3a and O3b are exploratory item-level estimates. Higher O7 values indicate greater perceived rule-violation or cheating risk. Panel D applies the prespecified O4 equivalence test to the same estimate and HC2 standard error. Bounds are ± 0.15 times the observed O4 SD among A1 respondents in the O4 analytic sample. *p_L* and *p_U* are one-sided tests against the lower and upper bounds; equivalence requires both below .05, equivalently a 90% confidence interval wholly within the bounds. Equivalence was not established; this is inconclusive, not evidence of a substantively meaningful difference.

Appendix O: Robustness and Treatment-by-Background Analyses

Table O.1 reports the primary O1 estimate and post hoc robustness checks, followed by preregistered exploratory treatment-by-background analyses for prior gaokao and public-sector examination participation. Table O.2 reports a compact post hoc ordinal sensitivity analysis for O1.

Table O.1. Robustness and Treatment-by-Background Analyses for O1

Analysis or subgroup	<i>N</i>	<i>N</i> _{A1}	<i>N</i> _{A2}	Estimate (HC2 SE)	95% CI	<i>p</i>
<i>Panel A. Primary estimate and post hoc robustness checks</i>						
Primary unadjusted estimate	2,410	1,204	1,206	-0.151 (0.044)	[-0.237, -0.066]	< .001
Complete-case sample: unadjusted	2,402	1,200	1,202	-0.151 (0.044)	[-0.237, -0.066]	< .001
Complete-case sample: adjusted	2,402	1,200	1,202	-0.153 (0.044)	[-0.238, -0.067]	< .001
Gaokao-taker restricted sample	2,348	1,172	1,176	-0.148 (0.044)	[-0.235, -0.062]	< .001
Public-sector-exam-taker restricted sample	1,240	612	628	-0.176 (0.061)	[-0.295, -0.057]	.004
<i>Panel B. Prior gaokao participation (D6): preregistered exploratory model</i>						
Did not participate	58	30	28	-0.155 (0.288)	[-0.720, 0.410]	.591
Participated	2,348	1,172	1,176	-0.148 (0.044)	[-0.235, -0.062]	< .001
Interaction: participants minus nonparticipants	2,406	1,202	1,204	0.006 (0.291)	[-0.565, 0.578]	.982
<i>Panel C. Prior public-sector examination participation (D8): preregistered exploratory model</i>						
Did not participate	1,149	582	567	-0.121 (0.063)	[-0.245, 0.002]	.055
Participated	1,240	612	628	-0.176 (0.061)	[-0.295, -0.057]	.004
Interaction: participants minus nonparticipants	2,389	1,194	1,195	-0.055 (0.088)	[-0.226, 0.117]	.531

Notes: A1 denotes the nationally uniform-paper condition and A2 the province-specific-paper condition. Except for interaction rows, entries are A2–A1 contrasts in O1; interaction rows compare those contrasts across participants and nonparticipants. The primary estimate is preregistered and confirmatory; the other Panel A estimates are post hoc. The two complete-case rows use the same sample, and the adjusted model includes categorical age group, gender, and education. The taker-only Panel A rows are separate restricted-sample models; the corresponding Panel B and C margins come from the preregistered D6 and D8 interaction models. Parentheses contain HC2 standard errors; 95% confidence intervals and *p*-values are two-sided. For interaction rows, *N* and arm counts refer to the full model. Because D6 and D8 were measured after assignment, subgroup patterns do not identify causal moderation; the interaction coefficient is the heterogeneity test. The D6 nonparticipant estimate is especially imprecise (*N* = 58).

Table O.2. Post Hoc Ordinal Sensitivity Analysis for O1

<i>Panel A. O1 response distributions</i>				
Response category		A1, <i>n</i> (%)	A2, <i>n</i> (%)	
1. Very unfair		69 (5.7)	70 (5.8)	
2. Relatively unfair		248 (20.6)	310 (25.7)	
3. Neither unfair nor fair		171 (14.2)	211 (17.5)	
4. Relatively fair		625 (51.9)	541 (44.9)	
5. Very fair		91 (7.6)	74 (6.1)	
<i>Panel B. Proportional-odds ordered-logit sensitivity</i>				
Contrast	<i>N</i>	Common odds ratio (robust 95% CI)	Robust <i>p</i>	PO-test <i>p</i>
A2 versus A1	2,410	0.751 [0.647, 0.872]	< .001	.198

Notes: Percentages in Panel A use each arm's nonmissing O1 responses. Panel B fits a proportional-odds ordered logit to O1 with A2 as the sole predictor and reports the exponentiated treatment coefficient with a sandwich-robust confidence interval and two-sided *p*-value. An odds ratio below one indicates lower odds of selecting a higher O1 category in A2. The proportional-odds (PO) restriction is assessed by a likelihood-ratio comparison with a nonparallel cumulative-logit model; its *p*-value does not indicate a detectable violation. This post hoc exploratory sensitivity analysis does not replace the preregistered OLS mean contrast.

Appendix P: Post-Treatment Diagnostics and Post Hoc Analyses

Table P.1 reports the preregistered post-treatment diagnostics. Table P.2 reports post hoc descriptive contrasts and measurement diagnostics, including the selected-sample O1 contrasts in Panel C. Table P.3 inventories the post hoc additions, deviations, and diagnostics. Figure P.1 reports completion time.

Table P.1. Treatment-Reception, Information-Equivalence, and Data-Quality Diagnostics

Panel A. Correct-response and pass rates					
Diagnostic	A1 (%)	A2 (%)	Difference (HC2 SE), pp	95% CI, pp	p
O9a correct response	95.8%	96.6%	0.8 (0.8)	[-0.7, 2.4]	.286
O9b correct response	90.5%	70.5%	-20.0 (1.6)	[-23.1, -17.0]	< .001
O9c correct response	93.4%	94.5%	1.2 (1.0)	[-0.7, 3.1]	.230
O9d correct response	96.0%	95.4%	-0.6 (0.8)	[-2.2, 1.0]	.484
AC1 pass rate	99.4%	99.6%	—	—	—
Panel B. Full response distributions					
Response	A1, n (%)		A2, n (%)		
O9a					
All provinces use the same examination paper	1,155 (95.8%)		24 (2.0%)		
Different provinces use different examination papers	36 (3.0%)		1,166 (96.6%)		
The material did not mention this	6 (0.5%)		7 (0.6%)		
Not sure / did not notice	9 (0.7%)		10 (0.8%)		
O9b					
Common syllabus and the same maximum score	1,092 (90.5%)		851 (70.5%)		
Different syllabi and the same maximum score	40 (3.3%)		202 (16.7%)		
Common syllabus and different maximum scores	24 (2.0%)		62 (5.1%)		
The material did not mention this	26 (2.2%)		64 (5.3%)		
Not sure / did not notice	24 (2.0%)		28 (2.3%)		
O9c					
Quotas allocated across provinces, then selection within province	1,126 (93.4%)		1,141 (94.5%)		
Applicants compete nationally for all positions	38 (3.2%)		21 (1.7%)		
Employing units determine their own quotas	2 (0.2%)		5 (0.4%)		
The material did not mention this	24 (2.0%)		22 (1.8%)		
Not sure / did not notice	16 (1.3%)		18 (1.5%)		
O9d					
Selection in descending order of examination scores	1,158 (96.0%)		1,152 (95.4%)		
Selection mainly from application materials	9 (0.7%)		4 (0.3%)		
Selection by random lottery	2 (0.2%)		0 (0.0%)		
The material did not mention this	22 (1.8%)		18 (1.5%)		
Not sure / did not notice	15 (1.2%)		33 (2.7%)		
AC1					
Supermarket	0 (0.0%)		0 (0.0%)		
Bank	5 (0.4%)		4 (0.3%)		
Post office	1,199 (99.4%)		1,202 (99.6%)		
Library	2 (0.2%)		0 (0.0%)		
Hospital	0 (0.0%)		0 (0.0%)		
Not sure	0 (0.0%)		1 (0.1%)		

Notes: Panel A reports correct-response rates for O9a–O9d and pass rates for AC1. O9 differences are A2 minus A1 in percentage points, with HC2 standard errors in parentheses, 95% confidence intervals, and two-sided *p*-values. AC1 is summarized descriptively because no between-arm inferential test was prespecified. O9a assesses reception of the assigned paper arrangement; O9b–O9d assess reception of features held constant in the written vignettes. Panel B reports response counts and percentages of all completed records in each arm. None of these items has missing responses. All Table P.1 diagnostics were prespecified and did not enter the preregistered outcome analyses. Table P.2, Panel C reports the post hoc descriptive O1 contrasts for the four O9-correct subsets.

Table P.2. Post Hoc Descriptive Contrasts and Measurement Diagnostics

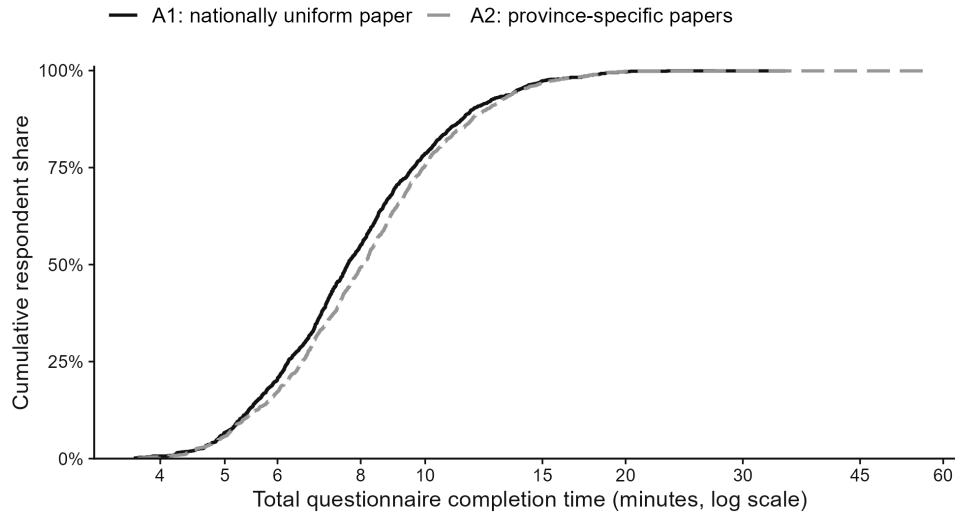
<i>Panel A. Within-respondent O8-minus-O5 contrast</i>										
Analysis	<i>N</i>			A1 mean	A2 mean	Difference (HC2 SE)	95% CI	<i>p</i>		
O8 minus O5	2,391			0.485	0.111	-0.374 (0.041)	[-0.454, -0.293]	< .001		
<i>Panel B. O3 two-item relationship</i>										
Analysis	Complete pairs, <i>N</i>			Pearson's <i>r</i>	Spearman's ρ	Spearman–Brown reliability				
Pooled	2,411			0.785		0.773	0.880			
A1	1,205			0.717		0.675	0.835			
A2	1,206			0.645		0.538	0.784			
<i>Panel C. O1 contrasts within post-treatment correct-response subsets</i>										
Correct-response subset	Sample size			Mean O1		Difference (HC2 SE)	95% CI	Raw <i>p</i>	Holm <i>p</i>	
	<i>N</i>	<i>N</i> _{A1}	<i>N</i> _{A2}	A1	A2					
O9a correct response	2,318	1,153	1,165	3.337	3.189	-0.148 (0.045)	[-0.235, -0.060]	< .001	.004	
O9b correct response	1,941	1,090	851	3.340	3.222	-0.118 (0.049)	[-0.214, -0.022]	.016	.016	
O9c correct response	2,265	1,125	1,140	3.335	3.199	-0.136 (0.045)	[-0.224, -0.048]	.002	.005	
O9d correct response	2,308	1,157	1,151	3.336	3.191	-0.145 (0.044)	[-0.232, -0.058]	.001	.004	

Notes: All three panels are post hoc. A1 and A2 denote the nationally uniform-paper and province-specific-paper conditions, respectively. Panel A reports arm means of the respondent-level O8–O5 score among complete pairs and their A2–A1 difference. A negative value means that, relative to A1, A2 reduced perceived paper-arrangement fairness more than perceived quota-allocation fairness; it does not measure respondents' relative valuation of the two dimensions. Panel B reports pooled and arm-specific relationships between O3a and O3b; both items are positively related within each arm. These summaries do not establish measurement invariance, and the pooled relationship combines within-arm association with the treatment-induced between-arm mean difference. They did not alter the preregistered complete-pair O3 index. Panel C reports A2–A1 mean differences in O1 in four item-specific samples of respondents with observed O1 who answered the named O9 item correctly. Because O9 followed assignment, these comparisons do not preserve randomized comparability and are descriptive, not compliance-adjusted causal effects, robustness checks, or validation analyses. These selected-sample contrasts depart from the PAP's restriction against constructing estimates conditional on post-treatment diagnostic responses. They were added after inspection of the formal results to describe the O1 pattern among respondents who correctly answered each O9 item. The preregistered O1 specification and analytic sample remain unchanged. Parentheses report HC2 standard errors; 95% confidence intervals and raw *p*-values are two-sided, and Holm *p*-values adjust jointly across the four Panel C contrasts.

Table P.3. Post Hoc Additions, Deviations, and Diagnostics

Analysis	Status	Reported in	Interpretive scope
Complete-case reference and compact covariate adjustment	Addition	Table O.1	Separates the complete-case sample change from covariate adjustment; neither model replaces the primary unadjusted O1 estimate.
Gaokao-taker restricted model	Addition	Table O.1	Restricted-sample robustness; does not replace the D6 interaction.
Public-sector-exam-taker restricted model	Addition	Table O.1	Restricted-sample robustness; does not replace the D8 interaction.
O8-minus-O5 paired contrast	Addition	Figure 6 and Table P.2	Compares treatment penalties across two fairness dimensions; does not measure which dimension respondents care about more.
Completion-time ECDF	Addition	Figure P.1	Post-treatment process measure; never used for sample restriction.
Age-16 hukou province map	Addition	Figure M.1	Descriptive geographic coverage; not a representativeness claim.
O1/O7 completed-sample item-missingness diagnostics	Addition	Table M.2, Panel B	Extreme-value bounds and an O7 item-nonresponse diagnostic apply only to completed questionnaires; they are not confidence intervals or an attrition analysis.
O1 ordered-outcome sensitivity analysis	Addition	Table O.2	One post hoc proportional-odds model for O1; does not replace the preregistered difference in mean ratings.
O1 contrasts within O9 correct-response subsets	Deviation	Table P.2, Panel C	Selected-sample diagnostic added after formal-result inspection; see the table notes for the PAP departure. The preregistered O1 analysis is unchanged.
O3 item-relationship diagnostic	Addition	Table P.2	Scale-construction diagnostic only; the preregistered complete-pair O3 index is unchanged.

Notes: All entries were developed after preregistration and initial inspection of the formal results. The O9-correct contrasts depart from the PAP's restriction on conditioning estimates on post-treatment diagnostics; their selected-sample interpretation is explained in Table P.2.

Figure P.1. Distribution of Total Questionnaire Completion Time by Treatment Arm

Notes: Lines show empirical cumulative distributions for all 2,413 completed records. Median completion time was 7.65 minutes in the nationally uniform-paper arm and 8.05 minutes in the province-specific-paper arm.